

Intrinsically disordered protein

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Proteins can exist in a trinity of structures: the ordered state, the molten globule, and the random coil. The five following examples suggest that native protein structure can correspond to any of the three states (not just the ordered state) and that protein function can arise from any of the three states and their transitions. (1) In a process that likely mimics infection, fd phage converts from the ordered into the disordered molten globular state. (2) Nucleosome hyperacetylation is crucial to DNA replication and transcription; this chemical modification greatly increases the net negative charge of the nucleosome core particle. We propose that the increased charge imbalance promotes its conversion to a much less rigid form. (3) Clusterin contains an ordered domain and also a native molten globular region. The molten globular domain likely functions as a proteinaceous detergent for cell remodeling and removal of apoptotic debris. (4) In a critical signaling event, a helix in calcineurin becomes bound and surrounded by calmodulin, thereby turning on calcineurin's serine/threonine phosphatase activity. Locating the calcineurin helix within a region of disorder is essential for enabling calmodulin to surround its target upon binding. (5) Calsequestrin regulates calcium levels in the sarcoplasmic reticulum by binding approximately 50 ions/molecule. Disordered polyanion tails at the carboxy terminus bind many of these calcium ions, perhaps without adopting a unique structure. In addition to these

examples, we will discuss 16 more proteins with native disorder. These disordered regions include molecular recognition domains, protein folding inhibitors, flexible linkers, entropic springs, entropic clocks, and entropic bristles. Motivated by such examples of intrinsic disorder, we are studying the relationships between amino acid sequence and order/disorder, and from this information we are predicting intrinsic order/disorder from amino acid sequence. The sequence–structure relationships indicate that disorder is an encoded property, and the predictions strongly suggest that proteins in nature are much richer in intrinsic disorder than are those in the Protein Data Bank. Recent predictions on 29 genomes indicate that proteins from eucaryotes apparently have more intrinsic disorder than those from either bacteria or archaea, with typically >30% of eucaryotic proteins having disordered regions of length ≥ 50 consecutive residues. © 2001 by Elsevier Science Inc.

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INTRODUCTION

The 20th century saw the development and solidification of the protein structure–function paradigm, which can be represented in brief as:

Amino-Acid Sequence $\rightarrow$ *3Dimensional Structure* $\rightarrow$ *Function*.

The key point of this paradigm is that 3D structure is taken to be an obligatory prerequisite for protein function, so native protein structure equates with ordered 3D structure. In this introduction, a

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brief history of this paradigm is presented, followed by information that is leading to a call for its reassessment.

A Brief History of the Current Protein Structure–Function Paradigm

Extracellular extracts of beer yeast (containing invertase) hydrolyzed α -glucosides but not β -glucosides, while emulsin hydrolyzed the latter but not the former. From these observations, in 1894 Fischer¹ wrote (as translated in Lemieux and Spohr²), “To use a picture, I would like to say that enzyme and glucoside have to fit to each other like a lock and key in order to exert a chemical effect on each other.”

In 1936, Mirsky and Pauling³ compiled the following information: loss of pepsin activity correlated with the amount of protein denatured⁴; acid, alkali, and urea all increased the viscosity of protein solutions and denatured the proteins without aggregation⁵; many native proteins form characteristic crystals while denatured proteins don’t crystallize; exposure of sulfhydryls and other side-chain groups and denaturation typically occur concomitantly. From this information and other observations, they concluded, “The characteristic specific properties of native proteins we attribute to their uniquely defined configurations. The denatured protein molecule we consider to be characterized by the absence of a uniquely defined configuration.” Slightly earlier than Mirsky and Pauling, Hsien Wu followed a series of twelve papers on protein denaturation with a thirteenth,⁶ which probably contains the first statement⁷ that protein function depends on prior structure. Neither Mirsky and Pauling nor Wu cited the earlier work of Fischer, whose lock-and-key proposal provided a very strong argument supporting their main thesis.

The next decade saw increasing numbers of experiments on protein denaturation. These experiments added to the list of proteins, supporting the view that function depended on 3D structure.⁸ In 1952 Edsall speculated: “Is it reasonable to hope that the endless variety of proteins found in nature, and their extraordinary diverse and specific interactions with one another and with other substances, can be explained on the basis of a few relatively rigid general patterns, simply by varying the nature and sequence of the side chains attached to the fundamental repeating pattern?”⁹ This idea took root and came to underlie almost all of the subsequent work and thinking.¹⁰

By the time RNase had been renatured *in vitro*^{11,12} and by the time atomic resolution structures of myoglobin¹³ and lysozyme¹⁴ had been determined, a highly specific 3D structure was already long accepted as the necessary prerequisite for protein function. The subsequent avalanche of more than 12,000 structures¹⁵ has buried alternatives to the current protein structure–function paradigm, yet most of these structures are similar to each other and have recognizable sequence similarity to only a small fraction of the proteins in nature.^{16–19}

Evidence for the Lack of Generality of the Current Structure–Function Paradigm

In 1950, Karush²⁰ reported that, unlike essentially every other native protein known at the time, serum albumin exhibited a nearly universal capacity for the high-affinity binding of small, hydrophobic, typically anionic molecules. Competitive binding was demonstrated for molecules of very different shapes. Using the same logic as Fischer for the lock and key, Karush inferred that albumin’s binding site assumes a large number of

configurations of similar energies in equilibrium with each other, and that, upon interacting with a given hydrophobic anion, the best-fitting configuration becomes selected from albumin’s structural ensemble. Karush called this phenomenon configurational adaptability. Subsequent studies on albumin binding support Karush’s original proposal of an ensemble of structures, but with some added interesting twists, such as the existence of different conformers with slow rates of conversion, corresponding to slightly less than 15 seconds.²¹

Amino acid side chains differ from one to another but all are joined by the same peptide bond within proteins. This indicates the need for conformational changes during protein synthesis to accommodate the differently shaped side chains. From this insight, Koshland independently proposed configurational adaptability, but instead named this process induced fit.²² Configurational adaptability and induced fit are evidently the first suggestions that significant conformational change is responsible for protein function. At the time induced fit was proposed, no mechanism was implied, leaving open the question of whether the process of binding induces a new conformation or involves selection of the best-fit alternative from an ensemble of structures in equilibrium.

Evidence supporting a significant domain movement upon substrate binding was first presented for the association of glucose with hexokinase in 1978.²³ The bound and free enzyme structures were of different isozymes due to failure to obtain crystals of the same isozyme with and without ligand. The inferred domain shifts were further supported by large shape changes upon binding in solution as measured by small-angle X-ray scattering.²⁴ This domain shift upon ligand binding was taken as support for the induced-fit hypothesis and used to explain the astonishing differences between ATPases and kinases. The latter have the specific need to prevent water from reacting at the active site; the domain movements were proposed to decrease water accessibility.²⁵ Thus, the induced-fit hypothesis shifted from a mechanism for binding different substrates to a way of gaining deeper understanding of domain movements upon ligand binding.

More recently, aspartate aminotransferase, aromatic amino acid transferase, and especially a chimera of the two were shown to be capable of operating on a diverse set of substrates by means of structural accommodations within the binding pocket.²⁶ More than any of the domain-shifting examples, this protein fits Koshland’s original description of induced fit as a means for acting on a set of related but structurally distinct molecules. However, due to the appropriation of induced fit for explaining domain movements upon binding, the authors suggested multi-induced fit as a way to explain their data.

Oxygen binding by hemoglobin and catalysis by aspartate transcarbamoylase are both regulated by the binding of nonsubstrate ligands. For both proteins, nonsubstrate-ligand binding is associated with shifts between alternative 3D structures in multi-subunit proteins. Several models have been developed to explain this behavior, two of which are symmetric allostery, by Monod et al.²⁷ and sequential allostery, by Koshland et al.²⁸

Discovery of Intrinsically Disordered Protein Segments

In contrast to the view that function depends strictly on prior 3D structure, or on structural accommodations within a prior 3D structure, or on regulatory shifts between alternate structures, researchers discovered examples in which nonstructured segments of proteins play important roles in protein function.

Indeed, more than 20 years ago, at a time when only about 20 protein crystal structures had been determined, some protein segments were discovered to yield no discernable electron density and yet to be essential for function.^{29,30} A possible relationship between these studies and Karush's much earlier insights²⁰ has not been considered until now.

Missing electron density in protein structures can arise from failure to solve the phase problem, from crystal defects,³¹ or even from unintentional proteolytic removal during protein purification. However, a common reason for missing electron density is that the unobserved atom, side chain, residue, or region fails to scatter X-rays coherently due to variation in position from one protein to the next, i.e., the unobserved atoms are disordered. The literature now contains numerous reports of disordered regions that are crucial for function.^{32–36}

In 1978, the same year that functional disorder was indicated by X-ray crystallography, NMR revealed the highly charged, functional tail of histone H5 to be disordered.³⁷ Since then, NMR 3D structural determination has led to the characterization of several proteins containing functional, yet disordered regions.^{38,39} A further, surprising result of NMR protein structure determinations was the discovery of functional proteins that apparently lack any discernable structure, i.e., apparently native proteins that are disordered from end-to-end.^{40–42} Because NMR is more certain in its characterization of disorder than X-ray diffraction, the rediscovery of native disorder by NMR had significant impact.^{43,44}

The Two-State Model of Protein Folding

Denaturation by urea or guanidine can cause globular proteins to unfold from their compact shapes into extended forms often called random coils in deference to earlier work in polymer chemistry. Protein random coils differ from true random coils in important respects, such as a nonrandom population of internal bond angles along the chain. A variety of data supported a two-state model for protein folding. In such models, denaturation, involves just the ordered and random coil states without significant populations of any intermediate forms.^{8,12}

Partially Folded Intermediates

In contrast to the two-state view of protein folding and denaturation, partially unfolded intermediates between the ordered state and the random coil have been observed as the major species in urea, guanidine, and pH titration studies for several (but not all) proteins.^{45–47} These folding intermediates exhibit side chains with motional characteristics more like those of the random coil but with backbone secondary structure more like that of the ordered state. A surprise at the time was the discovery that these folding intermediates were compact, with only slight expansion compared with their ordered states, as initially determined by intrinsic viscosity⁴⁸ and later confirmed by dynamic light scattering^{49,50} and small angle X-ray scattering.^{50,51}

Ptitsyn et al. provided an interpretation for the partially unfolded state.⁴⁸ In their model, the protein converts from an ordered (they used the term “native”) state into a form having some liquidlike characteristics, for example with the side chains going from rigid to nonrigid packing, while the secondary structure remains almost unchanged and the shape remains compact. Furthermore, Ptitsyn and Crane-Robinson coined the term “molten

globule”⁵² to describe the model for the liquidlike, partially folded state.

There has been considerable uncertainty regarding the molten globule hypothesis.⁵³ For example, it is uncertain whether heat, guanidine, urea, and pH extremes induce the same folding intermediate as was originally claimed.⁴⁸ For bulk liquids the properties are uniform, and, except for rare examples, like superfluidity, that depend on macroscopic quantum states, transitions from one type of liquid to a second type of liquid are not observed. In contrast to the behavior of liquids, partially folded α -lactalbumin⁵⁴ displays a series of related forms and apomyoglobin⁵⁵ reveals two distinct partially folded intermediates that appear sequentially during refolding. Some of the amide groups in partially folded α -lactalbumin show substantial protection against hydrogen exchange.⁵⁶ Indeed, as reported at this meeting, folding intermediates of a very small protein, bovine pancreatic trypsin inhibitor, exhibit both a less mobile core and a more mobile disordered region as determined by hydrogen exchange.⁵⁷ Such nonuniform protection against exchange is inconsistent with the original molten globule model,⁵⁸ which included a proposal for rapid structural fluctuations. In addition, confusion arose because researchers were not careful in distinguishing between actual protein behavior and Ptitsyn's idealized molten globular model.⁵⁸ By now, however, many of the critics have largely come to accept the term molten globule, if not all of the details of the original proposal.^{59,60}

Of course, a wide range of behaviors should be expected for molten globules as a result of variations in structure and sequence between different proteins. Furthermore, the existence of more than one liquid phase can be reconciled with Ptitsyn's molten globule hypothesis by noting that the properties of liquids become granular as the volume becomes sufficiently small. Perhaps because of their small size and sequence heterogeneity, proteins (unlike bulk liquids) can evidently exhibit more than one liquidlike (molten globular) form. With these caveats in mind, we agree with Ptitsyn that the molten globule represents a third thermodynamic state for proteins.⁶¹

The molten globule was initially discovered as an equilibrium structure observed in studies on protein denaturation. Later this protein form was proposed to exist as a transient intermediate during the kinetics of protein folding.⁵⁴ Indeed, several ingenious studies have provided evidence for this proposal, at least for some proteins.⁶² Besides existing as a stable intermediate during *in vitro* denaturation and as a transient during *in vitro* folding of some proteins, the molten globule has also been proposed to provide the basis for biological function *in vivo*. The insertion of proteins into membranes⁶³ and the transfer of retinal from its bloodstream carrier to its cell-surface receptor^{64,65} have both been suggested to depend on the molten globular state. These and other examples support the existence of the molten globule form *in vivo*.⁶⁶ Additional proteins that form molten globules rather than ordered structure under physiological conditions continue to be discovered.^{67–72}

The information presented above suggests that it is appropriate to reassess the current protein structure/function paradigm. The remainder of this review attempts such a reassessment.

INTRINSIC DISORDER

This section discusses briefly the methods used to characterize intrinsic disorder and presents The Protein Trinity. This hypothesis relates native protein function to the three thermodynamic states of protein and to their transitions. Next, in support

of The Protein Trinity, we present five proteins that use disorder and, for some, transitions between order and disorder, for their functions. We then present 16 more examples grouped as historically important examples, wholly unfolded proteins, and disordered regions with interesting functions. Next, our efforts to understand sequence–disorder relationships and our predictions of disorder from sequence are briefly reviewed. Protease digestion is sometimes used to argue against the existence of intrinsic disorder, so this argument is explored along with the role of chaperones. We end with a brief comment on the implications of intrinsic disorder for the structural genomics project.

Determination of Intrinsic Disorder

Several methods have been used to characterize intrinsic disorder in proteins, each having its own strong and weak points.

X-ray crystallography: As mentioned above, disorder leads to missing electron density in protein structures determined by X-ray crystallography. Two types of disorder have been recognized: static and dynamic.^{31,73} If a dynamic region freezes into a single preferred structure upon cooling, then collecting data at lower temperatures distinguishes dynamic from static disorder in some cases.⁷⁴ However, from our point of view, more important than static or dynamic, is whether the missing region assumes one set of the Ramachandran ϕ , ψ angles along the backbone or whether the missing region exists as an ensemble of angles. We call a missing region with one set of ϕ , ψ angles, whether static or dynamic, a “wobbly domain” because such a region assumes different positions as a rigid body, with the transitions between the different positions being slow (static disorder) or fast (dynamic disorder) on the X-ray analysis time scale. From our point of view, a region existing as an ensemble of ϕ , ψ angles, whether static or dynamic, is intrinsically disordered. The major uncertainty regarding information from X-ray diffraction is that, without additional experiments, it is unclear whether a region of missing electron density is a wobbly domain, is intrinsically disordered, or is the result of technical difficulties.

NMR spectroscopy: 3D structures can be determined for proteins in solution by NMR. The absence of the requirement for crystallization means that NMR provides a less biased estimate of the commonness of disorder compared with crystallography. Under favorable circumstances, NMR provides motional information on a residue-by-residue basis⁷⁵ by means of a variety of different isotopic labeling and pulse sequence experiments (for an excellent review, see the paper by Braken in this volume⁷⁶). Especially useful is the ^{15}N - ^1H heteronuclear NOE measurement, which gives positive values for (more slowly tumbling) ordered residues and negative values for (more rapidly tumbling) disordered residues.^{75–77} When these NOE data are displayed for an amino-acid sequence, ordered regions are evident as a series of consecutive positive values and disordered regions as consecutive negative ones.³⁸

Compared with ordered proteins, relatively few molten globules have been structurally characterized by NMR, indicating the existence of significant experimental difficulties. First, proteins with molten globular regions often aggregate at the concentrations needed for NMR. Second, molten globules are heterogeneous with structural interconversions on the millisecond timescale; this leads to extreme broadening of the side-chain NMR peaks. Indeed, to obtain side-chain information for molten globules, fitting

to molecular simulations data⁷⁸ and residue-specific isotopic labeling⁷⁹ were used in place of the more standard approaches. Also, NMR backbone data on molten globules are also difficult to obtain due to the opposite effects, that is because of a lack of chemical shift dispersion (rather than extreme line broadening).

Unlike most other molten globules, the molten globule formed by apomyoglobin at pH 4 is stable up to 50°C. At this temperature, the individual peaks sharpen enough so that even poorly dispersed spectra can be used.⁸⁰ Employing this strategy, Wright et al. were able to determine a backbone structure for the pH 4 apomyoglobin molten globule, revealing regions of secondary structure and flexibility.⁸¹ Although NMR is not biased against random-coil disorder, the molten globule presents experimental difficulties that can be overcome only with great difficulty if at all. Thus, native molten globular protein domains would probably be underrepresented in any collection of disordered proteins characterized by NMR.

Circular dichroism (CD) spectroscopy: Structural information for proteins in solution is also provided by circular dichroism.⁸² Far-UV CD spectra provide estimates of secondary structure and so distinguish ordered and molten globular forms from random coil. On the other hand, near-UV CD show sharp peaks for aromatic groups when the protein is ordered, but these peaks disappear for molten globules and random coils due to motional averaging.^{48,52,83} Thus, combined use of near and far-UV CD can distinguish whether a protein is ordered, molten globular, or random coil. However, this method is only semi-quantitative and lacks residue-specific information and so does not provide clear information for proteins that contain both ordered and disordered regions.

Protease digestion: By the late 1940s, it was already recognized that protease digestion gave insight into protein structure and flexibility,⁸⁴ with a more detailed view emerging after more had been learned about protein structure.⁸⁵ More recent studies by Fontana et al.^{86–89} provide compelling evidence that flexibility, not mere surface exposure, is the major determinant for digestion of possible cut sites.

Since a structured region has to become unfolded over more than 10 residues to be cut by typical proteases,⁸⁸ the ratio of the digestion rates, (disordered)/(ordered), should be about equal to $e^{-\Delta G/RT}$, where ΔG is the free energy of unfolding the segment. For two-state unfolding, whole protein unfolding free energies and segmental unfolding free energies would not be distinguishable. For typical proteins, unfolding free energies range from 7 to 10 Kcal/mol,⁹⁰ so a disordered region would be expected to undergo digestion $\sim 10^5$ to 10^7 times faster than an ordered one.

Studies by Fontana et al. demonstrate huge increases in digestion rates after the F helix of myoglobin is converted to a disordered state in apomyoglobin, with the cut loci for several different proteases occurring within the disordered region that arises from the F helix. The exact (disorder)/(order) digestion rate ratio was not determined, but the authors indicate⁸⁷ that the relative rates could be even larger than the 10^5 to 10^7 estimated above. Thus, hypersensitivity to proteases is sure evidence of protein disorder. Protease digestion gives position-specific information. However, the requirement for protease-sensitive residues limits the demarcation of order/disorder boundaries by this method.

Protease digestion is especially useful when used in combination with other methods. For example, protease digestion can be used with X-ray diffraction to help sort out whether a region of

missing electron density is due to a wobbly domain or to intrinsic disorder.^{91,92} A protein with two structured domains connected by a short flexible linker converts to two lower molecular-weight fragments with no digestion intermediates, while a protein with intrinsic disorder typically exhibits a set of intermediate-sized fragments (from random cleavage of the intrinsic disorder) that converge to homogeneous fragments at longer times of digestion. Protease digestion is also useful when coupled with CD spectra, which lack position-specific information. Finally, the combination of proteolysis and mass spectrometry for fragment identification shows special promise for indicating the presence of intrinsic disorder.⁹³

Stoke's radius determination: Random-coil disorder has also been detected by various methods for obtaining Stoke's radius such as small-angle X-ray scattering or size exclusion chromatography.⁹⁴ Abnormally large radii for a given molecular weight provide evidence of disorder. Such methods have been used in combination with CD spectroscopy,^{95,96} which provides independent evidence of random coil structure. If the random coil is responsible for function, then protein activity is not lost by incubation at high temperatures.⁹⁶

The Protein Trinity

As stated above, the current protein structure/function paradigm emphasizes the role of ordered 3D structure as being a necessary prerequisite to protein function. Our alternative proposal, herein called The Protein Trinity, is shown in Figure 1. In this view, native, intracellular proteins or functional regions of such proteins can exist in any one of the three thermodynamic states noted above⁶¹: ordered forms, molten globules, and random coils.

The key point of The Protein Trinity is that a particular function might depend on any one of these states or a transition between two of the states. In this view, not just the ordered state, but any of the three states can be the native state of a protein.

These three states can be viewed as analogous to solid (ordered), liquid (molten globule) and gas (random coil) and their

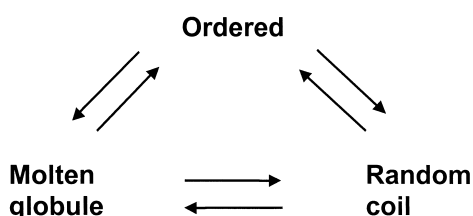
interconversions are analogous to phase transitions. In this analogy, the term "disordered" would be equated with the term "fluid" used as an adjective, encompassing both the molten globule and the random coil.

The transitions between the ordered state and the random coil and between the ordered state and the molten globule exhibit easily measurable cooperativity.^{97,98} However, given the small sizes of proteins, the heterogeneity of the internal side-chain packing, and the irregularities of the backbone structures, the transitions are not nearly so sharp as for bulk phases.

Liquids and gases usually exhibit-clear phase transitions whereas the transitions between the molten globule and the random coil typically do not.^{61,97} However, some artificial, synthetic helical bundles exhibit both molten globule-like, nonrigid side-chain packing and cooperative unfolding.^{99,100} The behavioral differences between these artificial bundles and natural molten globules appear to rest in differences in the two types of hydrophobic cores: the artificial bundles were constructed to have purely hydrophobic side chains in their cores while actual proteins in the molten globule state have mixtures of both polar and nonpolar side chains in their cores.¹⁰⁰ Thus, the presence or absence of a cooperative transition depends on the details. The Protein Trinity analogy to solid/liquid/gas can still hold by considering the molten globule $\leftrightarrow$ random coil transitions without discernable cooperativity as analogous to liquid $\leftrightarrow$ gas transitions at temperatures and pressures above their critical points.

Disorder and flexibility are often used synonymously, but the two terms are quite distinct. With regard to an ordered protein, flexibility refers to the magnitudes of the excursions of the atoms from their equilibrium positions. The Debye-Waller temperature factors (also called B-factors) obtained from X-ray data provide experimental measures of this sort of flexibility.^{101,102} In addition, molecular dynamics was invented to study of this type of motion.¹⁰³ Indeed, almost countless numbers of papers have been published that use molecular dynamics simulations to explore local flexibility. For a disordered region, variation in flexibility refers to differences in the speed of interconversion among the various members of the structural ensemble. A variety of methods have been used to investigate the flexibilities of disordered regions and proteins, including NMR as discussed previously⁷⁵ and in this volume.⁷⁶

THE PROTEIN TRINITY



Proposal: Function can arise from any of the three protein forms and transitions between them.

Figure 1. An alternative hypothesis for protein structure/function. As shown, The Protein Trinity suggests that native protein structure includes not only the ordered state, but also random coils and molten globules. Further, this hypothesis suggests that function can be carried out, not only by the native state, but by any of the three states or their transitions.

Examples in Support of The Protein Trinity

The formulation of The Protein Trinity proposal was derived from observations showing that numerous protein functions depend on nonordered protein forms and also that some functions depend on transitions between two forms. Here we discuss five such proteins and their structural transitions that we have studied in our own laboratories (Table 1).

Filamentous phage membrane insertion: The fd phage comprises 2,700 almost entirely helical coat proteins of 50 residues each. The subunit helices lie approximately parallel to the fiber axis arranged in layers with five subunits per layer. Each helix has a slight curvature that winds around the long axis of the phage. Subunits overlapping like scales on a fish lead to the formation of a helical bundle more than 8,500 Å long. During infection, the coat protein helices somehow slide relative to each other while penetrating the inner membrane of the host *E. coli*. This movement eliminates the subunit overlap in the capsid so that

Table 1. Five Examples of Functional Disorder

Protein	Disorder	Function
fd Phage	ordered → molten globule	Membrane penetration
Histone octamer	ordered → molten globule	DNA replication, Transcription, Chromosome remodeling
Clusterin	Native molten globule	Protein detergent
Calcineurin	95 aa disordered loop	Display of calmodlin target
Calsequestrin	21 aa disordered tail	Ca ⁺⁺ binding

the subunits come to lie at a common level in the cell membrane. As first characterized by Griffith et al.,^{104–107} a chloroform–water interface converts the filaments into shortened rods called I-forms at lower temperatures and then the I-forms convert into spheroidal structures called S-forms at slightly higher temperatures (Figure 2). An especially important aspect of this figure is the irregularity

of I-forms and S-forms when observed by typical electron microscopy, but their greater regularity when observed in the hydrated state. Regularity was observed previously for the I-forms and S-forms if they were cross-linked with glutaraldehyde before examination by electron microscopy.¹⁰⁵ The I-forms are proposed to mimic the intermediate step as the coat proteins insert into the

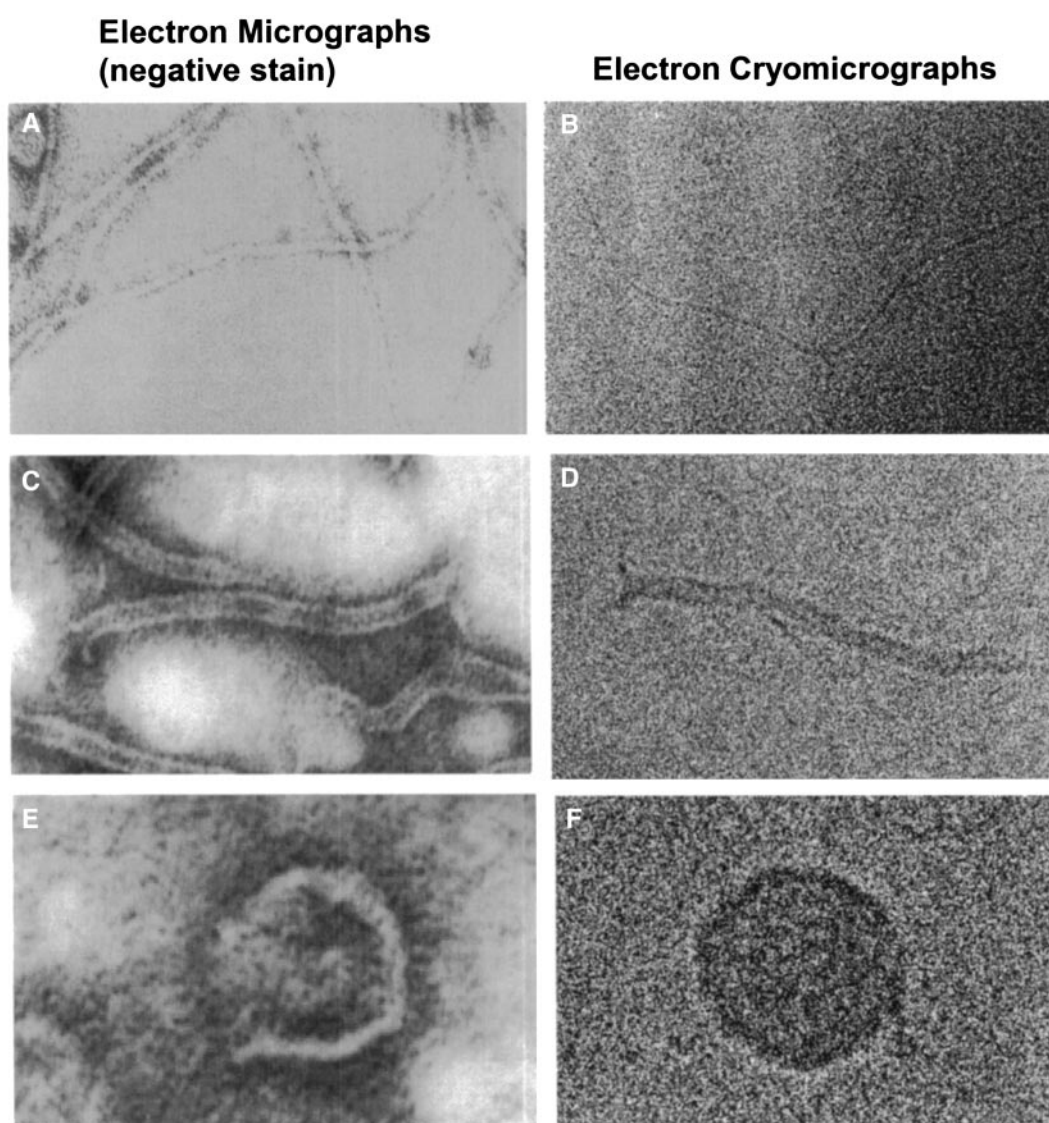


Figure 2. Contraction of filamentous phage by a chloroform/water interface. Images of fd filaments (a, b), I-forms (c, d) and S-forms (e, f) were prepared using either traditional negative stain electron microscopy (a, c, e) or using cryo-electron microscopy without stain (b, d, f).

host membranes, whereas the S-forms, which have a morphology resembling vesicles yet contain no lipid, are proposed to mimic the final, membrane-inserted forms.^{108,109} Three lines of evidence suggest that both I-forms and S-forms, but not the filaments, have whole-particle molten globular characteristics: (1) The I-forms and S-forms, but not the filaments, exhibit substantial fragility when subjected to conditions for electron microscopy¹¹⁰ (Figure 2). (2) The tryptophan fluorescence of I-forms and S-forms is extraordinarily sensitive to nonpolar quenchers but normally sensitive to polar quenchers, while the tryptophan fluorescence of filaments is normally sensitive to both nonpolar and polar quenchers. These data suggest that the nonpolar quenchers dissolve into the hydrophobic interiors of the I-forms and S-forms due to nonrigid side-chain packing, but do not dissolve into the interiors of the filaments due to tight packing.¹⁰⁸ (3) I-forms and S-forms, but not filaments, associate with the polarity-sensitive probe, 1-anilino-8-naphthalene sulfonate (ANS), to greatly enhance the extrinsic fluorescence of this molecule. These data show that this probe can associate with the hydrophobic interiors of the I-forms and S-forms (as it does with known molten globules) due to nonrigid side-chain packing, but that this probe does not associate with the interiors of the filaments due to tight packing.¹¹⁰

The loose side-chain packing of such intermediate forms would enable the subunits to slide relative to each other as required for the phage infection process.¹⁰⁹ Of course the fd phage provides an interesting example of the earlier suggestion that the molten globule offers an ideal structure for the insertion of membrane proteins.⁶³

Crushing typical and hyperacetylated nucleosomes: The disk-shaped nucleosome core particle contains a histone octamer wrapped by 1 3/4 turns of double-stranded DNA having 146 base pairs in total. The histone proteins contain positively charged tails that are sensitive to protease digestion¹¹¹ and that are unobserved in the nucleosome X-ray structure,^{112,113} both of which concur in the assignment of these tails to the disordered class of protein structure.

Hyperacetylation of these tails was discovered more than 30 years ago and has been implicated in both DNA replication^{114,115} and RNA transcription.^{114–116} Solution studies comparing typical and hyperacetylated nucleosomes discern small or no changes in structure or stability due to this modification.^{117,118} On the other hand, electron micrographs show that hyperacetylated nucleosomes become highly deformed on the carbon surface whereas normal nucleosomes maintain their normal shape. This indicates that hyperacetylation causes nucleosomes to become much more sensitive to the forces encountered during sample preparation,¹¹⁹ similar to the I-forms and S-forms of fd phage. Furthermore, hyperacetylation causes the negatively charged nucleosome to become even more negatively charged. Note that increased charge imbalance, induced for example by low or high pH, is one of the primary methods for promoting transitions from the ordered state to the molten globule.^{48,61,62} Ignoring bound and associated counter ions and assuming 1/2 charge unit for histidine due to its pKa near 7, we estimate a net charge of -138 for nucleosomes without acetylation, and up to as much as -164 for the most fully acetylated particles.

To test whether the increased charge repulsion leads to a decrease in the rigidity of nucleosomes, we turned to atomic force microscopy (AFM), which we have been using for similar investigations in fd phage.¹²⁰ The ability to control the amount of force on the particles is one of the major advantages of this method.

Another is that samples can be studied in a hydrated state at atmospheric pressure (not in a vacuum as for typical electron microscopy) or even under water.

AFM images developed using tapping mode revealed very little difference between normal (Figure 3A, left) and hyperacetylated nucleosomes (Figure 3A, right). Contact mode, but not tapping mode, can be used to vary the force between the tip and the surface, so a series of experiments were performed using this protocol. Normal nucleosome images taken at ~ 47 nN (Figure 3B, left) and retaken at the same force after exposure to 110 nN (Figure 3B, right) showed very little change. On the other hand, at even lower forces, ~ 29 nN, images of hyperacetylated nucleosomes showed considerable distortion (Figure 3C, left). Furthermore, following exposure to 38 nN and reimaging at the lower ~ 29 nN force, the hyperacetylated nucleosomes were largely destroyed (Figure 3C, right). Note especially that the region subjected to ~ 38 nN is covered with irregular matter that most likely corresponds to debris from crushed nucleosomes.

A large number of experiments like that shown in Figure 3 were performed using various sample preparations and several tips having different flexibilities. The resulting curves of measured height versus force for normal and hyperacetylated nucleosomes show the hyperacetylated nucleosomes to be crushed by less than 40 nN while bulk nucleosomes remain intact out to 275 nN (Figure 4). Thus, hyperacetylation makes nucleosome core particles less rigid.

Clusterin, a protein with a native molten globular domain:

Enrico Sertoli¹²¹ first described cells, now bearing his name, that are found in close association with developing germ cells within the testis. By now it is apparent that Sertoli cells “nurse” the germ cells by providing nutrients and regulatory factors.¹²² One protein, called clusterin, was found as a secretion from Sertoli cells and is presumed to play an essential role in germ cell development.¹²³ By sequence matching, this protein was determined to be the same as other proteins found later in various tissues, especially the brain.¹²⁴ Finally, this same protein was also identified as being associated with cellular injury, lipid transport, and apoptosis.¹²⁵

Structural characterization of clusterin has been difficult. Repeated attempts at crystallization have failed. This protein forms aggregates and sticks to hydrophobic surfaces. Although there are several possible explanations for such behavior, one is the presence of a molten globular domain.

We are developing methods for characterizing proteins that might contain both ordered and molten globular domains in the same structure. Given its behavior, clusterin provides a good model system for testing our approach. The methods focus on using protease digestion to distinguish ordered and disordered regions of a protein and ANS fluorescence to determine whether any protease-sensitive regions are molten globules or random coils. In addition, as discussed below in more detail, we are developing several predictors of natural disorder regions (PONDRs) that predict order and disorder from amino acid sequence.^{126,127} In essence, these predictors compare a given sequence with the set of structurally characterized ordered and disordered sequences used for training.

Protease digestion of native and denatured clusterin and predictions of order and disorder by PONDR show almost perfect agreement: in the native state, regions predicted to be disordered are sensitive to protease digestion, while regions predicted to be ordered are resistant to digestion. The resistant regions become sensitive to digestion when clusterin is in a nonnative state (Figure

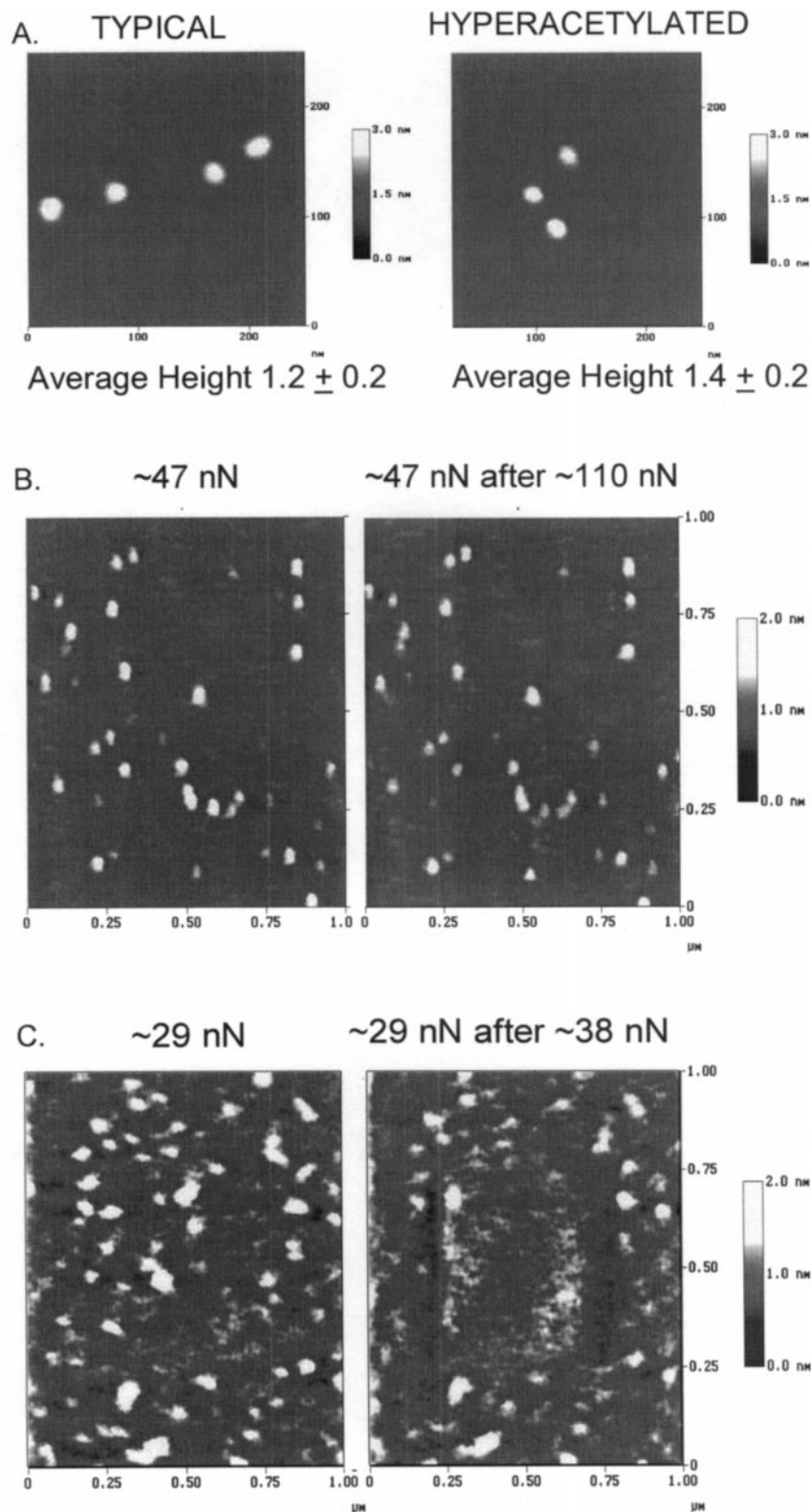


Figure 3. Atomic force microscopy of typical and hyperacetylated nucleosomes. (A) Typical (left) and hyperacetylated nucleosomes (right) are shown for AFM images prepared using tapping mode. (B) Image of typical nucleosomes using contact mode with a tip force of ~ 47 nN (left) and an image for the same sample using the same mode at the same tip force, but after the central region had been subjected to a tip force of ~ 110 nN (right). (C) Image of hyperacetylated nucleosomes using contact mode with a tip force of ~ 29 nN (left) and an image for the same sample using the same mode at the same tip force, but after the central region of the image had been subjected to a tip force of ~ 38 nN (right). Note that only nucleosome debris is visible in the central region in (C).

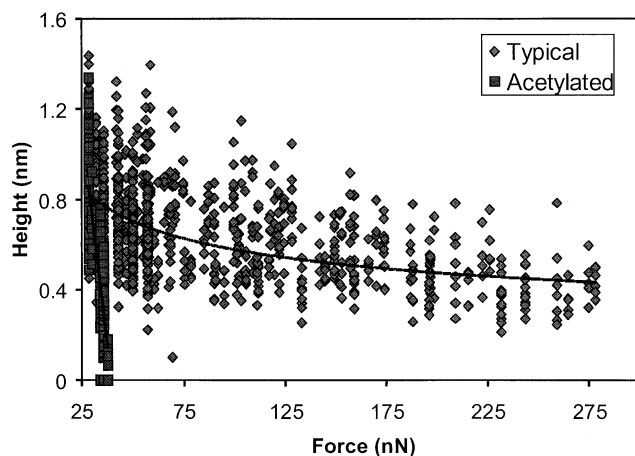


Figure 4. Height versus force for typical and hyperacetylated nucleosomes. Typical and hyperacetylated nucleosomes were imaged using AFM in the contact mode at various force values. To span such a wide range of forces, 4 different tips with different degrees of stiffness were used.

5). Furthermore, ANS binds to clusterin as indicated by 100-fold increases in extrinsic fluorescence (data not shown). Also indicative of ANS binding is the observation that ANS inhibits trypsin digestion at one or possibly two of the sites located within the protease-sensitive, disordered region (Figure 5). The ANS binding and its protection of particular sites from trypsin digestion suggest that the disordered regions in clusterin very likely form molten globular structure.

The function of clusterin is unclear, but recent unpublished data show that this protein dissolves bacteriorhodopsin and thus acts as a proteinaceous detergent. Such a function is commensurate with the properties of the molten globular state. Furthermore, a detergentlike function could be important whenever cellular components need to be moved, such as for cellular remodeling in sperm

maturation, in building projections from the cellular bodies in brain tissue, and in removing cell debris following apoptosis.

Calcineurin, a protein with essential disorder: This protein contains a catalytic A subunit and a calmodulin-like B subunit (Figure 6A). This model was constructed from coordinates obtained from PDB (accession code: 1au1) as originally determined by Kissinger et al.⁹² The calcineurin A subunit is a calcium/calmodulin-activated serine/threonine phosphatase.¹²⁸ Being involved in both calcium and phosphorylation signaling pathways, calcineurin plays important roles in many if not all eucaryotic cells. For example, calcineurin was so-named because of its high concentrations in brain tissue,¹²⁹ yet this same protein plays a key role in T-cell mitogenesis. With respect to the latter, inhibition of calcineurin's phosphatase activity is the key event in immunosuppression by drugs such as cyclosporin and FK506, which are commonly used to prevent rejection following organ transplantation.^{130,131} Disorder plays an important role in calcineurin function. The calmodulin target helix is located within a 95-amino-acid disordered segment that is unobserved in crystal structures.⁹² The lack of ordered structure in this region was previously discovered due to its hypersensitivity to protease digestion.⁹¹ Calmodulin completely surrounds its helical target upon binding¹³² as shown in Figure 6B (the coordinates for this figure were taken from PDB accession code: 1cdm). Locating this helical target within a region of disorder enables it be easily surrounded by calmodulin so the region of disorder is essential to the regulation of calcineurin by calcium/calmodulin.

Calsequestrin, a calcium storage protein: Calcium release out of the sarcoplasmic reticulum (SR) stimulates muscle contraction while calcium uptake into the SR leads to muscle relaxation. A protein within the lumen of the SR facilitates the calcium uptake/release functions by high-capacity, low-affinity calcium binding. This protein, calsequestrin, binds 40–50 calcium ions with affinities of ~ 1 mM.^{133–135} Rabbit skeletal muscle calsequestrin has three regions of missing electron density in its crystal structure¹³⁶: a glutamate dimer (EE) at the amino terminus; a

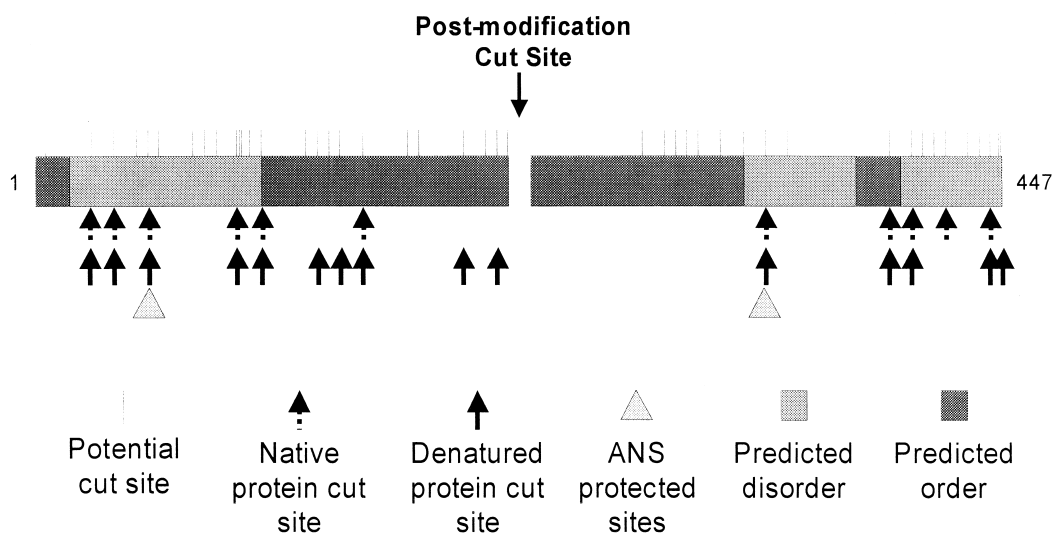


Figure 5. Protease digestion of clusterin. Purified clusterin with and without renaturation was subjected to digestion by trypsin and the fragments were separated by HPLC. The earliest appearing fragments correspond to the most sensitive cut sites, which were identified by amino-acid sequence analysis. Protection against trypsin digestion by ANS is weak (if the protection exists at all) for the site on the left, while the protection is clear and unambiguous for the site on the right.

seven-residue disordered loop with five consecutive negative charges (MDDEEDL) from residue 327 to 333; and a 20-residue carboxy terminus having 16 glutamates + aspartates (EGEINT-EDDDDEDDDDDDDD), and thus having 17 negative charges when the carboxy terminus is included. There are barely enough excess negative charges in calsequestrin to bind to the ~50 calcium ions sequestered by this protein. Thus, all of the disordered, negatively charged residues are almost certainly involved in calcium binding.

A calsequestrin tetramer from the crystal structure¹³⁶ is shown in Figure 6C. The coordinates used for the ordered parts of the protein in this figure can be found in PDB (accession code: 1a8y). Please note that the structural representations of the disordered regions (yellow) are completely ad hoc. The amino terminus and 327–333 loop each represent a hypothetical single conformation of what must be a large ensemble of native protein conformations. Furthermore, the 20-residue carboxy-terminal tail certainly does not adopt the conformation shown. The purpose of showing this polyanion tail in a highly extended conformation is to demonstrate that the tail has many options for calcium-mediated interactions with other negative side chains (in red), including possible calcium cross bridges with each other and also with the seven-residue disordered loop.

Implications for The Protein Trinity

Disorder-to-order transitions upon binding are becoming recognized as a common occurrence (see examples below), with the ordered-state endpoint being crucial for function. For such proteins, it might be considered a minor detail whether the ordered state occurs before, or concomitant with, molecular association. The five examples given above do not follow the pattern of disorder-to-order upon binding; indeed, for these examples, an ordered state may have little or nothing to do with function.

For our first two examples above, the protein assemblies undergo order-to-disorder transitions (not the reverse) and the resulting disordered state is responsible for function. For the filamentous phage, the nonrigid side-chain packing allows the subunits to move relative to each other. This facilitates the sliding motion that must occur as the coat protein penetrates the cell membrane. For the nucleosome core particle, reducing particle rigidity would facilitate local structural deformations, thereby allowing access to the DNA without a requirement for dissociating the histones from the DNA. How DNA replication and RNA transcription can occur without separating the histones from the DNA has been a great mystery for both of these processes.

In our third example, calmodulin completely wraps around its target helix in calcineurin upon binding. The target helix must be well separated from the remainder of the protein (Figure 6B). The ~20-residue target helix is located within a 95-residue disordered region. The disorder allows the target helix to separate from the rest of the protein, thus permitting access for binding. For the calmodulin/target helix interaction, it is unknown and possibly unimportant whether the other ~75 disordered residues become ordered upon binding. The important function of disorder in this example is to display and make accessible the target helix. Not only does trypsin digestion rapidly activate calcineurin, thereby confirming the flexibility of this disordered region, but trypsin digestion activates a number of other calmodulin-regulated enzymes as well,⁹¹ suggesting that disorder adjacent to the calmodulin target helix is a common feature of these enzymes.

In our final examples, clusterin and calsequestrin both have disordered regions that bind ligands. The former contains a molten globular domain that apparently functions as a proteinaceous detergent and the latter has several negatively charged disordered segments that bind calcium. For both proteins, ligand binding does not necessarily bring about a specific disorder-to-order transition. The tails of detergents typically do not become ordered when incorporating hydrophobic molecules into their micelles; perhaps the same lack of induction of order occurs as well when the molten globular clusterin domain associates with its ligand. As shown in Figure 6C, the polyanion tail of calsequestrin can potentially come close to a number of carboxyl groups at very different locations on the protein surface; thus, assuming that the tails are random coils in the absence of calcium (not the extended structures shown in this figure to illustrate their potential reach), addition of calcium ions could induce many different possible cross bridges leading to many different, likely static structures. Thus, calcium binding by calsequestrin need not correspond to the induction of a particular structure, but rather most probably induces an ensemble of protein structures, most or all members of which exhibit low affinity calcium binding.

More Examples Supporting The Protein Trinity

There have been numerous reports of proteins that contain regions of intrinsic disorder required for function. Here we discuss examples selected for their historical importance and potential interest (Table 2).

Trypsinogen: More than 20 years ago, studies on this protein pointed to the importance of protein disorder.^{30,137} Indeed, this is the first high-resolution protein crystal structure that identified a functionally important region of disorder by its missing electron density, with two independent structure determinations in basic concurrence.^{30,138} In both of these studies, residues 1 to ~15 were unobserved.

Every biochemical textbook explains that proteolytic digestion removes the VDDDDK amino terminus and that this cleavage converts inactive trypsinogen into active trypsin; none that we can find explains that the site of cleavage is within an ~15-residue region with intrinsic disorder, the flexibility of which is important. The structural changes that occur after the cleavage to convert inactive trypsinogen into active trypsin are less clear and in some dispute.^{138,139} One side in this dispute suggests that the structure of trypsin's substrate-binding pocket, although exhibiting some disorder, can be resolved on the basis of a single structure that is the predominate one in the ensemble.¹³⁸ Researchers on the other side suggests that this region, which they call the "activation domain," is disordered.³⁰ Several additional experiments support the latter interpretation as discussed in detail by Bennett and Huber.¹³⁹ If the latter is true, then cleavage of the amino terminus brings about a disorder → order transition in the substrate-binding pocket.

The proteolytic cleavage of trypsinogen's amino terminal VD-DDDK sequence creates a new amino terminus in trypsin. The amino terminal VD dipeptide in trypsinogen becomes the more hydrophobic amino terminal IV dipeptide in trypsin. According to Huber et al., this new, hydrophobic dipeptide inserts into its own binding site, thereby stabilizing the correctly folded activation domain. Indeed, added IV dipeptide can bind to uncleaved trypsinogen and stabilize the activation domain in the structure that it adopts in trypsin.³⁰ These data suggest that the function of the last six residues of the disordered tail is to inhibit proper

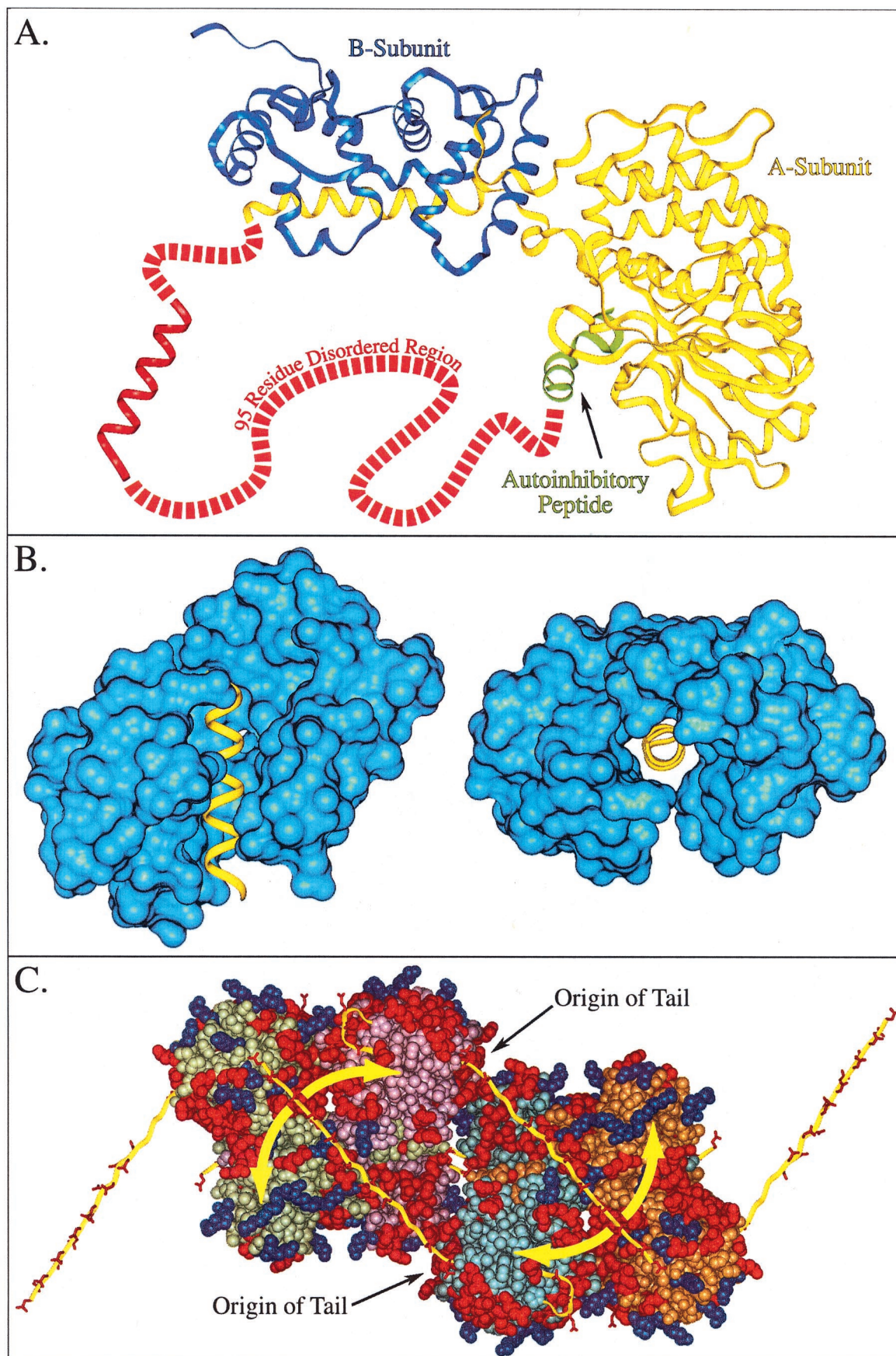


Figure 6. Structures of proteins with essential disorder. (A) The A subunit of calcineurin contains a phosphatase domain (yellow), a helix (yellow extension) that binds the B subunit and an autoinhibitory peptide (green) that binds to the active

Table 2. Important Examples of Native Disorder

Protein	Disorder Length	Function
Trypsinogen	~15	Folding inhibitor
TMV capsid	25	RNA binding
Lac repressor	61	DNA binding
Calmodulin	4	Flexible linker, Target binding
p21 ^{Waf1/Cip1/Sdi1}	164	Kinase inhibitor
HIV-1 gp120 V3 loop	24	Cell attachment

folding by capping a critical dipeptide terminus and that the remaining 12 disordered residues allow the cleavage-created IV terminus to search for and find its binding site.

Tobacco mosaic virus (TMV) coat protein: The TMV capsid is a rod-like structure containing more than 2,000 coat-protein molecules and an RNA of 6,400 nucleotides long.¹⁴⁰ Reports on this protein were the second to identify a functionally important disordered region by its missing electron density.^{29,141,142} Furthermore, NMR experiments indicate dynamic rather than static disorder for this region.¹⁴³

The TMV coat protein's disordered region is a 25-residue loop that is positively charged. This loop undergoes a disorder-to-order transition upon RNA binding during the assembly of the capsid. As pointed out by Holmes,¹⁴⁰ the flexibility and disorder of this region is absolutely essential for the observed assembly pathway. That is, more than 30 TMV coat-protein molecules associate to form a rodlet. The RNA joins this preformed protein rodlet by threading into a hole in its center. Without the flexibility of the region of disorder, there would not be enough room in the central hole to thread the RNA.

Lac repressor, a paradigm for gene regulation: This protein represents a prototype for understanding gene regulation. It binds to a specific DNA sequence and thereby prevents transcription of the associated operon.¹⁴⁴ The DNA binding domain comprises a 61-amino-acid segment at the amino terminus that is unobserved in the crystal structure of the protein, but this segment is observed and bound to DNA in protein/nucleic acid co-crystals.³⁵ NMR studies on DNA binding by the first 56 amino acids of this fragment indicate local regions of structure, but also highly mobile loops that become immobilized upon binding to DNA.¹⁴⁵ Thus, like the TMV capsid protein, the lac repressor undergoes a disorder-to-order transition upon binding to nucleic acid.

Calmodulin, a protein with two disorder features: Calmodulin has two globular domains, each with 2 EF hands that bind calcium. In the protein crystal, a helix connects these two globular domains, yielding the famous dumbbell structure.¹⁴⁶ The connecting helix is an artifact of crystallization. When calmodulin is in solution, part of the helical rod melts into a flexible linker,¹⁴⁷ thus enabling calmodulin to wrap around its target (Figure 6B).

Disorder plays a second role in this protein. Even after being saturated with calcium, the globular domains in calmodulin switch among several alternative conformations in solution, more so than many other folded proteins (Minji Zhang, personal communication). This limited disorder may account for the significant plasticity observed within the globular domains when they bind to different targets.^{148,149} This plasticity is reflected in alterations in the helix-helix contacts within the globular domains.¹⁵⁰ That is, the binding of target helices with different sequences probably selects out different members of an ensemble of structures as suggested by Karush for the binding of hydrophobic anions by the albumins.²⁰ Furthermore, the side chains in the binding surface have an abnormally high flexibility,¹⁵¹ perhaps due to high proportions of methionine.¹⁵² The absence of branching permits methionine to assume many different shapes, thereby enabling this side chain to adopt many different conformations and so adapt to different hydrophobic surfaces. In this regard, we noticed that a high proportion of methionine has been observed in two completely different proteins that bind to a variety of hydrophobic ligands.¹⁵⁰

Cyclin-dependent kinase inhibitor, p21^{Waf1/Cip1/Sdi1}: This 164-residue protein binds to cyclin-dependent kinases, thereby inhibiting their activity. The unbound inhibitor appears to be completely disordered as indicated by protease digestion, CD, and NMR.⁴⁰ Upon binding to CDK2, the amino-terminal 84 amino acids adopt an ordered conformation. Thus, this is another example of a disorder-to-order transition, in this case, upon binding to another protein. The p21 molecule is an important component of cell-cycle control through its ability to inhibit many different cyclin-dependent kinases. The disordered state of native p21 is suggested to enhance this protein's ability to bind to multiple protein targets.⁴⁰

V3 loop of HIV-1 gp120: The gp120 protein from HIV-1 is involved in binding to receptors on host cells. Deletion of the V3 loop of this protein leads to loss of infectivity. A 24-residue fragment of the V3 loop of HIV-1 strain IIIB occurs as structurally heterogeneous peptides as shown by NMR spectroscopy.¹⁵³ This and similar V3 loop fragments occur in different β -turn conformations when bound to different V3 antibodies.^{154,155} These authors suggest that the V3 loop either exists as an ensemble of

Figure 6. (continued) site on the phosphatase domain. The B subunit (blue) binds to the A subunit helical extension and resembles calmodulin. The 95 amino acid intrinsically disordered region of the A subunit connects the end of the helical extension (residue 374) with the end of the autoinhibitory peptide (residue 470). This region of disorder was identified by its lack of electron density and so this part of the protein is unobserved. (B) Calmodulin (blue) bound to the target helix from calmodulin-dependent protein kinase II (yellow) is shown from two orientations: looking from the side of the target helix (left) and looking down the target helix (right). (C) Representations of 4 calsequestrin molecules are shown with a slightly different color for each. The charged residues are indicated, with aspartates and glutamates in red, and with lysines, arginines and histidines in blue. The 3 disordered regions are rendered in yellow with their aspartates and glutamates in red. The disordered regions, especially the polyanion tail, were rendered so as to emphasize their possible extents, not to indicate realistic possibilities for their structures.

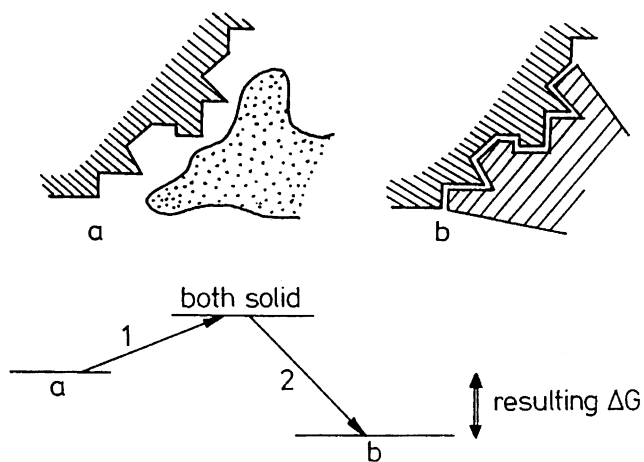


Figure 7. Schulz diagram. If a protein undergoes a disorder-to-order transition upon binding, the free energy to organize the disordered region is deducted from the free energies due to the contacts. This can potentially lead to high specificity coupled with low affinity.

structures with very different conformations or that binding forces conformational changes. The importance of the disorder-to-order transition of this loop is that a variety of chemokine receptors can be bound by this heterogeneous mixture of structures, allowing different avenues into the cell. This same mechanism makes devising a vaccine against HIV very difficult; some V3 loops can escape detection by antibodies that specifically recognize a given structure.

Interestingly, other pathogens have been shown to have attachment proteins that undergo disorder-to-order transitions upon binding to their receptors, including *Staphylococcal aureus*^{156,157} and Foot-and-Mouth Disease Virus.¹⁵⁸ Avoidance of the immune response¹⁵⁶ and ability to bind to different cell surface molecules¹⁵⁵ have both been suggested as possible adaptive advantages that derive from the local disorder in the unbound state.

Disorder and Molecular Recognition

The TMV capsid protein, the lac repressor, calmodulin, p21^{Waf1/Cip1/Sdi1}, and the HIV V3 loops all utilize disordered regions in molecular recognition via disorder-to-order transitions. From the lock-and-key perspective, the involvement of disorder in molecular recognition seems counterintuitive, yet by now a large number of similar examples have been discovered in addition to those in Table 2, suggesting that the involvement of disorder in molecular recognition is quite common. Here we discuss the thermodynamic and kinetic aspects of the involvement of intrinsic disorder in molecular recognition.

Thermodynamic aspects: To our knowledge, the first attempt to understand protein associations involving disorder-to-order transitions is the work of Schulz³² as shown in Figure 7. His simple figure, which we have named The Schulz Diagram¹⁵⁰ is a thought experiment, not a proposed reaction pathway. The first step (on the left) imagines making both partners rigid, which necessarily increases the free energy of the system; the second step (on the right) gives the sum of the free energies that arise as the result of the contacts between the two folded forms. Of course,

Table 3. Disorder and Molecular Recognition

Characteristic	Structural Variability
High Specificity/ Low Affinity	Flexibility in the Unbound State
High Affinity/ Low Specificity	Polymorphism in the Bound State

for many reactions, this sequence of events is simply impossible for steric reasons (e.g., see Figure 6B), which re-emphasizes that this figure does not represent a reaction pathway.

From this simple diagram, Schulz suggested that one possible advantage of disorder-to-order upon binding is the possibility of high specificity coupled with low affinity. As mentioned above, another possible advantage is the binding to multiple partners by structural accommodation as observed for p21^{Waf1/Cip1/Sdi1} (see Kriwacki et al.⁴⁰) and the HIV VP 3 loop.^{153,155} There is a certain symmetry for these two capabilities: the former depends on flexibility in the unbound state while the latter leads to structural polymorphism in the bound state¹⁵⁰ (Table 3).

The ability of one protein to bind to many partners has been called one-to-many signaling.¹⁵⁹ Calmodulin and p21^{Waf1/Cip1/Sdi1} are cases in point. Wright and Dyson⁴⁴ have recently reviewed intrinsically unstructured proteins with some emphasis on examples of this type. Disorder-to-order upon binding facilitates not only one-to-many signaling as for p21^{Waf1/Cip1/Sdi1} and calmodulin, but also many-to-one. An example is provided by interactions between nuclear localization signals (NLSs) and their receptor. The NLS does not depend on a precise sequence, but rather upon having a total of 4 lysines plus arginines within a segment of 6 amino acids that is flanked by negatively charged groups and helix breakers such as proline or glycine. The flanking regions help to insure “exposure” of the signal hexapeptide.¹⁶⁰ Intrinsic disorder would provide the need for exposure of the basic groups and, if long enough, would enable the NLS region to reach its receptor even if localized within the nuclear pore. An important feature of this system is the poor sequence conservation for the NLS regions, and yet these various regions apparently bind to the same receptor.

Disorder-to-order upon binding can overcome steric restrictions and thereby enable larger interaction surfaces in the complex than could be obtained for rigid partners. Two of many possible examples illustrate the point. First, as discussed above, a flexible linker allows calmodulin to almost completely surround its target helix¹³² as shown in Figure 6B. Second, multiple linkers between zinc finger domains (eight linkers between nine domains in the case of TFIIIA) enable such polydomain proteins to wrap up their DNA partners.^{161,162} When a protein undergoes a conformational change upon binding to a ligand, it is often unclear whether the change results from selection of one member of an ensemble by the best fit with the ligand, or whether the change results from induction of specific structure by interaction with the ligand. For special cases such as calmodulin–target helix interactions and TFIIIA–DNA interactions, steric clashes make it impossible to associate directly with one member from an ensemble, so the structure of the bound form must truly be induced by association with the ligand.

In addition to the qualitative reasoning discussed above, there have been studies aimed at making quantitative estimates of the free-energy contributions associated with disorder-to-order transi-

tions upon binding. The work by DeLisi et al.^{163,164} is especially germane to this problem. Of interest here are two sets of complexes that they studied: (1) a set of 9 serine endopeptidase-inhibitor complexes; and (2) a set of 13 flexible nonameric peptides interacting with the class I major histocompatibility complex (MHC) receptor. The first set involves association between basically rigid molecules while the second involves a disorder-to-order transition upon binding. Rather than attempting to estimate the binding free energies from first principles, DeLisi et al.¹⁶³ relieved some restrictions from an earlier semi-empirical method developed by Novotny.¹⁶⁵ In addition to the terms typically included in molecular mechanics energy functions, this method also includes terms for solvation and entropy associated with side-chain conformations, translational/rotational motion, and dilutional effects.

For the endoserine protease-inhibitor complexes, the observed and calculated values were within 2 kcal for 7 of the 9 complexes,¹⁶³ a result we find very impressive. The largest error, where the binding affinity is overestimated by 9.1 kcal, is for the trypsinogen-BPTI complex. The explanation for this large overestimate of course is in Figure 7. Because trypsinogen is disordered before binding³⁰ (e.g., see Table 2), the energy required for bringing about order, in this case perhaps ~ 9 kcal, takes away from the overall binding energy.

The results for the peptide-MHC receptor complexes are extremely interesting. To represent the ligand disorder, the authors created an ensemble of ligand structures using conformational searching. This ensemble was then analyzed to generate the various free energy terms for the receptor-ligand association. In their studies, DeLisi et al. don't specifically calculate the free energy change for the disordered ensemble converting an ordered state that is the same as the bound state but in the absence of receptor (thus corresponding to step 1 [left] in Figure 7). However, this free energy change can be estimated from their work by collecting terms dealing with the peptide only and leaving out all terms involving interactions between the peptide and MHC receptor, giving $\Delta G = E_1^b - E_1^f - T\Delta S_{bb}$ if the side-chain entropy contributions are ignored, and giving $\Delta G = E_1^b - E_1^f - T\Delta S_{bb} - T\Delta S_{sc}$ if the side-chain contributions are included, where the superscripts b and f mean bound and free, respectively, and subscripts bb and sc mean backbone and sidechains, respectively. When the values of the various terms estimated by DeLisi et al.¹⁶³ are plugged into these free energy equations, a slightly positive ΔG —when the side-chain contributions are ignored—becomes a very positive ΔG —when the side-chain contributions are included. Thus, these calculations add considerable insight into Figure 7, providing evidence that side-chain free energy contributions for disorder-to-order transitions are very important.

Some references to the work of DeLisi et al. conclude that the disorder-to-order transition upon binding leads to an increase in the overall binding affinity compared with the same binding by a rigid molecule. From the much simpler Schulz Diagram and from the analysis of the previous paragraph, a conclusion of increased affinity is simply incorrect.

Kinetic Aspects: The rate of association of a protein with its partner is generally expressed as $k_{\text{assoc}} \sim 4D\kappa f$, where D represents the relative translational diffusion coefficients and a is the sum of the radii (as in the Smoluchowski equation¹⁶⁶), and where κ and f are dimensionless parameters to account for orientational and electrostatic effects, respectively.¹⁶⁷ This leads to a modified Smoluchowski equation. The electrostatic

factor, f , can be very large. For example, the oppositely charged barnase and barstar exhibit association rates at high salt of about $10^5 \text{ M}^{-1}\text{s}^{-1}$ that increase to about $5 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$ at low salt.¹⁶⁸ In the absence of other effects, the orientational factor, κ , would be expected to be very small since the binding orientation represents a small fraction of all possible orientations.

Association rate constants frequently exceed the limits calculated from the modified Smoluchowski equation even when electrostatic effects are small. A model that explains the unexpectedly high association rates involves formation of an "encounter complex"¹⁶⁹ or a "transition state"¹⁷⁰ that can be thought of as an ensemble of conformations undergoing microcollisions. Even in the absence of attractive forces, Brownian motion predicts several microcollisions before the molecules diffuse apart.

Given the above framework for association rates, disorder can have several effects. First, the relative diffusion coefficient, D , would become smaller. Countering this effect, the effective size, a , would markedly increase. But the really large effects would occur for the orientational parameter, κ , where disorder could lead to productive collision for essentially any relative orientation of the two molecules. Thus, conversion from the ordered to the disordered state can lead to large increases in the rates of association.¹⁷¹ On the other hand, one could imagine cases in which the folding step becomes rate-limiting, in which case increasing the disorder would slow down rather than speed up the on-rates. The effects of disorder on dissociation rates are also complex. In cases where the two molecules cannot come apart because of steric restrictions, disorder would clearly have the effect of drastically speeding up the off-rate. The formation of disorder could also enable dissociation of part of the interface, like unzipping, which might also speed up dissociation.

Advantages of Disorder: From the above discussion, flexibility can either raise or lower binding affinities and either raise or lower on-rates (and most likely off-rates as well) compared with binding between rigid partners. Also, the binding affinities of rigid partners can be increased or decreased through mutation. Thus, for rigid partners, it surely would be possible to raise or lower specificity and to increase or decrease on- and off-rates via suitable mutations. Finally, as demonstrated by moonlighting proteins,¹⁷² rigid proteins can evolve to bind to more than one partner. These comparisons of the various properties describing molecular associations suggest that, despite earlier comments to the contrary, including some by us,^{150,159} there might be no intrinsic advantage for disorder-to-order transitions upon binding compared with association by rigid partners. The paper by Goldstein et al. presented at this meeting is especially germane to this point.¹⁷³ The argument is frequently made that metastable proteins exist because such instability is needed to carry out function. In contrast, Goldstein et al. argue that metastable proteins are favored during evolution because there are vastly larger numbers of sequences coding for these proteins compared with very rigid ones. Of course, as was pointed out in discussions at this meeting, functions that depend on intrinsically disordered proteins represent the extreme case of this argument. Thus, the involvement of intrinsic disorder in protein function may relate more to history and evolution than to advantage and necessity.

Native Proteins Without Secondary or Tertiary Structure

With the advent of structure-determination methods based on NMR, protein crystallization was no longer required to obtain a protein's 3D structure. To the surprise of nearly everyone, several proteins that carried out function in their respective *in vitro* assays were intrinsically disordered from end-to-end as revealed by NMR spectroscopy. Four of these wholly disordered, yet native proteins are discussed herein (Table 4). Before or concurrent with these NMR findings, several researchers had suggested the existence of native random coils on the basis of CD spectroscopy coupled with other methods;^{94,96,174} these CD-based studies failed to gain as much attention as the NMR-based work, in part because of the limitations associated with structural characterization by CD spectroscopy and in part because these CD-characterized, apparently native random coils lacked well-understood biological functions.

FlgM, a protein that regulates flagella assembly: NMR experiments indicate that FlgM is entirely disordered under physiological conditions and that, upon binding to σ^{28} , an anti-termination transcription factor, the carboxy-terminal half becomes ordered.⁴¹ This molecular interaction regulates flagella assembly in an interesting way. In *Salmonella typhimurium*, flagella are constructed from a transmembrane basal body attached to a polymer of the major protein subunit. More than 35 proteins assemble to form the basal body. In a puzzling regulatory mechanism, a mutation in any one of the basal body proteins shuts down transcription of the flagella subunit mRNA. This regulatory puzzle is solved by the FlgM protein.

FlgM is exported via a channel in the center of the basal body assembly, thus maintaining a low level of FlgM. However, if basal body assembly is defective as for any of the mutants, FlgM cannot exit via the channel and its levels rise. At these elevated levels, FlgM binds to σ^{28} and thereby shuts down synthesis of the mRNA for the flagella subunit. As flagella assembly nears completion, export of FlgM is also slowed due to the length of channel. This again shuts down synthesis of the flagella proteins.

FlgM almost certainly must be intrinsically disordered to migrate through the small diameter channel in the basal body assembly. If this scenario is correct, FlgM's regulatory mechanism depends on its own intrinsic disorder. This argument is reminiscent of the requirement for disorder in the TMV coat protein loop due to steric restrictions within the small central hole as originally discussed more than 15 years ago.¹⁴⁰

4E-binding protein 1 (4E-BP1): This protein, which contains 118 amino acids, inhibits translation by binding to the initiation factor eIF4E. NMR studies show 4E-BP1 to have little or no folded structure under physiological conditions,^{42,175} and yet this protein is able to inhibit translation in reticulocyte lysates under similar conditions. A short region in 4E-BP1, namely from residues 49 to 68, is responsible for binding to eIF4E.¹⁷⁶ This region undergoes a disorder-to-order transition upon binding to eIF4E, forming a helix followed by an extended loop on the surface of eIF4E.¹⁷⁷ Indeed, this region of 4E-BP1 acts as a molecular mimic of a helix plus loop in eIF4G, which is the normal docking site of eIF4E, thus explaining the competitive inhibition of the docking of eIF4E by 4E-BP1.

High Mobility Group Proteins I and Y, HMG-I(Y): These alternatively spliced transcripts from the HMG-I(Y) gene produce

Table 4. Wholly Unfolded Native Proteins

Protein	Disorder Length	Function
FlgM	97	Transcription promoter
4E-binding protein	118	Translation inhibitor
HMG-I(Y)	106	Architectural transcription factor
Neuromodulin	239	Calmodulin storage protein

the DNA-binding isoform proteins HMG-I(106 residues) and HMG-Y (95 residues). They were first identified by their high electrophoretic mobility among the nuclear proteins and are founding members of a new class of regulatory elements called "architectural transcription factors" that control the expression of a large number of human genes.¹⁷⁸ Various physical studies, including NMR spectroscopy, have demonstrated that, as free molecules in solution, the HMG-I(Y) proteins have no detectable secondary or tertiary structure. Discrete subregions of the proteins assume a planar, crescent-shaped configuration called the 'A.T-hook' when bound to the minor groove of A.T-rich DNA substrates.¹⁷⁹

Neuromodulin, a calmodulin storage protein, possibly involved in memory: Several similar proteins, including neurogranin and neuromodulin, are evidently unstructured proteins as determined by NMR.^{180–183} Calmodulin associates with these proteins at low, not high, calcium levels via a specific sequence called the IQ motif.^{182,184,185} In addition to calcium concentration, a second level of regulation in this system depends on phosphorylation.¹⁸⁶ Using membrane anchors for localization, neuromodulin/neurogranin may maintain calmodulin near the growing tips of the nerve cell projections when calcium levels drop. Because of this localization function, chains having the IQ motif have been called calmodulin storage proteins.¹⁸⁷

Interesting Functions Associated with Disorder

Although much of the focus on proteins centers on molecular recognition, protein molecules do have other functions that are important for maintaining life. The following examples present proteins with native disorder that have interesting functions not directly involving molecular recognition (Table 5).

The fd pIII protein: This phage protein, which is encoded by gene 3, is located on one tip of the fd filament and is responsible for attachment to the host *E. coli*.^{188,189} The existence and importance of a flexible linker for this protein was suggested long ago.^{190,191} Now we know that this protein contains three distinct domains: one integrated into one end of the filament and two that bind to a pair of co-receptors on the host cell.^{192,193} This double attachment is coordinated in an interesting way, suggesting the need for movement between the domains.^{194,195}

Two flexible linkers connect these three domains. The linker between the two attachment domains contains ~21 residues and the linker between the other two domains contains ~40 residues. Both are constructed from approximate repeats of the pentamer, EGGGS. As expected for such a sequence, the linker between the two attachment domains is unobserved in crystal structures.¹⁹⁶

The ability to easily remove the phage attachment domains by

Table 5. Native Disorder with Interesting Function

Protein	Disorder Length	Function
fd pIII	21, 40	Flexible linkers
Bcl-x _L	63	Display of protease and phosphorylation sites
Kinesin	~15	Stepping motor
Titin	2174	Entropic spring
K ⁺ Channel	64	Entropic clock
Neurofilament H	679	Entropic bristle

proteolysis¹⁹⁰ indicates that the conformation of the linkers is likely quite mobile. These linkers uncouple the two attachment domains from the body of the phage particle and allow motional freedom so that the attachment domains can reorient independently. This motional uncoupling speeds up searches for productive docking orientations between the attachment protein domains and their receptors. The flexible linker may also act as a shock absorber after attachment has occurred.¹⁹¹ A shock-absorber function has special importance in this system due to the extreme length of the phage.

Whenever a protein binds to two or more partners and the partners vary in spacing or orientation over time, just as discussed above for the fd attachment protein, flexible linkers can be essential. Thus, flexible linkers are quite common components of multifunctional enzymes, topoisomerases, transcription factors, and several other classes of proteins.^{161,197–204}

Bcl-x_L and its homologue, Bcl-2: These apoptosis inhibitors contain long, flexible loops characterized both by missing density in X-ray crystal structures and as being highly mobile by NMR in Bcl-x_L and characterized by similar location in the related protein, Bcl-2. In Bcl-2, this disordered loop plays a key role in programmed cell death.^{205,206} Furthermore, protease digestion^{207,208} and phosphorylation^{209,210} within the disordered loops of these two proteins play critical roles in modulating programmed cell death. We suggest that the intrinsic disorder of these loops plays an important role in the display of the protease digestion and phosphorylation sites. Each of these possibilities is discussed.

As discussed above, protease cut sites within disordered regions are digested orders of magnitude faster than potential cut sites within ordered regions. Molecular modeling provides insight into this hypersensitivity. Even though only 6 residues form close association with trypsin during proteolysis, molecular modeling indicates that a minimum of about 13 residues have to be unfolded for trypsin to be able to bind with high affinity. Without such local unfolding, binding is inhibited sterically.⁸⁸ In a folded protein, a 13-amino-acid segment would typically be folded nearly all the time and therefore would be inaccessible to digestion as discussed above. Thus, an unfolded local region of sequence (e.g., a disordered region) digests many orders of magnitude faster than when folded.⁸⁷

While trypsin binds six residues in the vicinity of the cleavage site, the cAMP-dependent protein kinase binds more than 20. The X-ray crystal structure of cAMP-dependent protein kinase shows the protein immobilizing a 20-residue peptide that spans the phosphorylation site.²¹¹ This peptide was derived from a natural 8-kD kinase inhibitor. Furthermore, fluorescence anisotropy measure-

ments on probes attached to site-directed cysteine mutants suggests a disorder-to-order transition as this peptide binds to the kinase.²¹² Other protein kinases appear to bind similarly large segments of their substrates. By analogy to the studies on trypsin, these results suggest that a phosphorylation site within a disordered region would react orders of magnitude faster than the same site within an ordered region. In support of this hypothesis, of the five phosphorylation sites structurally characterized before and after phosphorylation, four are disordered before phosphorylation and become ordered after the modification.²¹³

In addition to roles in proteolysis and phosphorylation, the disordered loop of Bcl-2 also appears to be involved in binding small molecules. The anticancer drug taxol induces apoptosis and this induction is overcome by overexpression of Bcl-2.^{214,215} Mobility shift experiments²¹⁶ and selection of taxol-binding peptides from a random library by phage display^{217,218} both suggest that taxol binds to the disordered loop of Bcl-2. To our knowledge, these are the first experiments demonstrating the binding of a drug to an intrinsically disordered region of a protein.

Kinesin: The kinesin family of microtubule-based motor proteins is involved in organelle transport, mitosis, meiosis, and several other cellular processes. As shown by Rice et al. using EPR spectroscopy and cryo-electron microscopy, a region linking kinesin's two 'head' domains undergoes a nucleotide- and microtubule-dependent disorder-to-order transition during the force-generating cycle.²¹⁹ More specifically, the disorder-to-order transition of the bound head's linker region positions the unbound head over the next binding site on the microtubule, thus facilitating the stepping motion of the motor.

The region in question appears as both ordered and disordered in different crystal structures.^{220–222} This structural variation is likely due to crystal packing forces as the various structures all have ADP bound in the active site. The corresponding region in myosin (the protein motor responsible for muscle contraction) also appears to undergo an analogous disorder-to-order transition during its force-generating cycle,^{223,224} although it is not clear exactly what role this transition plays.

The field of molecular motors has traditionally been approached using a mechanical analogy, i.e., the idea that a motor protein is just a very small assembly of levers, dashpots, and springs. However, it has recently become clear from the results of Rice et al.²¹⁹ as well as others,^{225,226} that disorder, dynamic flexibility, and even phase changes between order and disorder are important in both motor function and regulation. Thus, even for motor proteins, models incorporating thermodynamics (e.g., including phase transitions) may be more appropriate than models based solely on mechanical analogies.

Titin: This gigantic protein extends from mid-line to Z-line in muscle cells and is suggested to fix the length of the muscle cell. This protein has a region rich in P + E + V + K that, in some isoforms, is greater than 1,000 amino acids in length. The PEVK region is very likely to be structurally disordered and may function as a rubberlike entropic spring that helps to restore overstretched muscle cells to their natural, relaxed length.²²⁷

K⁺ Channel ball and chain: Voltage-gated K⁺ channels in nerve axons exist in three main states: closed, open and inactive. The closed state converts to the open state in response to an appropriate voltage gradient, while the inactive state is nonresponsive to voltage gradients. Conversion from the open state to the inactive state happens when a "ball" on the end of a flexible

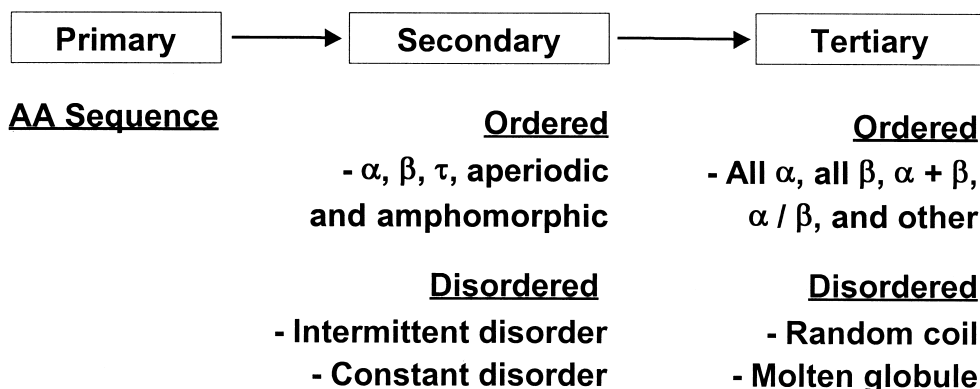


Figure 8. Implications for the protein structural hierarchy. The primary-secondary-tertiary structural hierarchy for proteins as found in all textbooks has been modified to include intrinsic disorder as a category of structure.

“chain” plugs the open channel.^{228–230} The timing of this closure is critical to proper nerve cell function and evidently depends on the length and flexibility of the chain. Here we suggest that this flexibility be recognized as an entropic clock due to its important function as a timing device.

Neurofilament H and M: Fibrous assemblies of neurofilaments L, H, and M are major components of motor neuron axons. The ends of the M and H filament proteins have long regions of disorder (>300 and >600 residues, respectively). These disordered ends were proposed to be entropic bristles that occupy space by thermally driven motion and thereby maintain the separation of neighboring filaments.²³¹ The separation of filaments also maintains the axonal bore, possibly allowing the movement of small molecules and maintaining the shape of the axon against compression. Numerous other proteins contain such disordered regions that are proposed to function as entropic bristles for modulating protein structure and function.²³²

Implications of Intrinsic Disorder for the Protein Structure Hierarchy

Why has it taken so long for concepts related to functional disorder to make it into the textbook world of biochemistry? First, there is an approximately 100-year history based on the lock-and-key model¹ and an approximately 70-year history equating native proteins with 3D structure.^{3,4} One term proposed to describe unfolded yet functional proteins, namely “natively denatured,”⁹⁴ is an oxymoron, but nevertheless serves to illustrate just how deeply ingrained is the perceived equivalence between ordered 3D structure and native protein structure. Indeed, in common speech, the terms “native” and “(ordered) structure” are used interchangeably.

In addition to “natively denatured,” the terms “natively unfolded”⁹⁶ and “intrinsically unstructured”⁴⁴ have been suggested to describe these proteins. These two terms seem to imply lack of any secondary structure and so would exclude native molten globules. We are therefore using the term “intrinsically disordered” to describe such proteins.

Structural ensembles in equilibrium (e.g., intrinsic disorder) were associated with protein function 50 years ago.²⁰ Furthermore, prominent examples such as activation of trypsinogen and DNA recognition by the lac repressor likely involve disordered

regions. Yet none of the widely used biochemical texts mention even a single example of the role of disorder in protein function.

In our view, a second contributor to the slowness in recognizing native protein disorder has been the current structural hierarchy taught in all the biochemistry texts: e.g., that protein structure can be explained as primary, secondary, tertiary, and quaternary. In this hierarchy, there is no category called intrinsic disorder. As has been discussed from the time of the early Greek philosophers, Plato and Aristotle, and as reviewed and discussed from a more modern perspective by Lakoff,²³³ categorization plays a key role in the development of knowledge. Since intrinsic disorder has not been included as a standard category of protein structure, textbook writers could not fit intrinsic disorder into their examples used to explain protein structure/function relationships.

The first three levels of the protein structural hierarchy (e.g., primary, secondary, tertiary) were developed to understand proteolytic digestion of proteins several years before any 3D structures were known.²³⁴ To our surprise, a formalized description of this hierarchy²³⁵ included a category of proteins called “partially folded.” Furthermore, about 30 years ago there were several interesting discussions of such partially unfolded proteins,^{32,33,137,143,236} but these concepts never made it into textbook biochemistry perhaps because of the rapid increase in the number of crystal structures being determined over the same time span.

Since intrinsic disorder and the transitions between protein states play important roles in native protein function, we have modified the first three steps of the current structural hierarchy (Figure 8). In this figure, τ is used for turns and amorphous is used for segments that switch between α and β structure, which may also represent a distinct category of secondary structure.²³⁷ The proposed additions in Figure 8 to include intrinsic disorder all involve time-dependent fluctuations or time-dependent changes in protein structure. Thus, Clare Woodward (discussion at this meeting) pointed out that these additions represent one way of adding the fourth dimension, time, to the currently static protein-structure hierarchy.

First, at the secondary structural level we suggest two categories of disorder: intermittent and constant. The former describes protein molecular recognition domains and other regions that undergo order/disorder transitions to carry out binding or other functions. The latter describes regions such as proteinaceous detergents, flexible linkers, entropic springs, entropic clocks, and

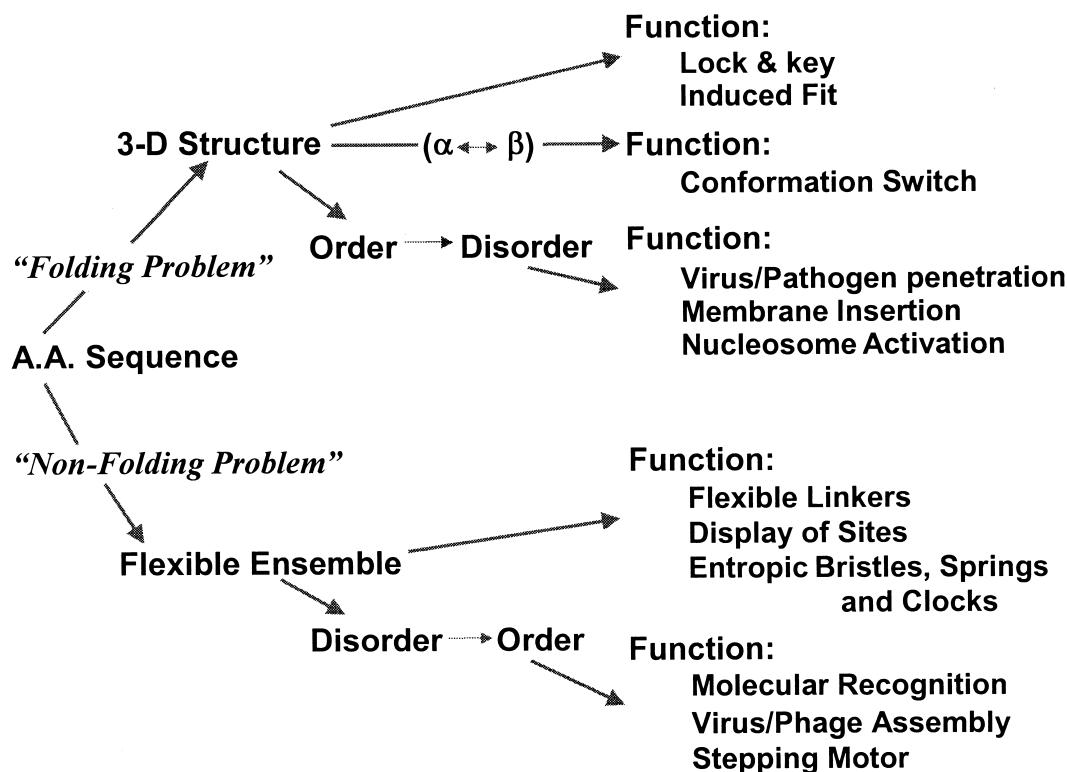


Figure 9. Protein structure/function relationships. The relationships presented in this figure are based on actual examples, many of which are given in Tables 1, 2, 4, and 5, but some of which have not been discussed in this review.

entropic bristles that remain disordered while carrying out function. At the tertiary level, we suggest two categories: random coil and molten globule. With this altered hierarchy, it becomes possible to explain protein structure/function – in terms of a diagram that includes not just *Amino-Acid Sequence* $\rightarrow$ *3D Structure* $\rightarrow$ *Function*, but also other structural forms and their transitions (Figure 9).

Figure 9 shows conformational switching, where a segment of protein switches from $\alpha \rightarrow \beta$ or from $\beta \rightarrow \alpha$; this idea is represented in Figure 8 by the term “amphomorphic.” Secondary structure switching has been observed for a long time.^{237–240} Recently, from a collection of such examples, a predictor of switch sequences was developed²⁴¹ and applied to myosin²⁴² in an effort to better understand the conformational changes in this molecule. Such conformational switching fits within The Protein Trinity’s solid/liquid/gas analogy. Many solids have different phases and cooperatively switch from one to another under appropriate conditions.

Also included in Figure 9 are the already-discussed transitions from order $\rightarrow$ disorder and from disorder $\rightarrow$ order. From the examples given above, these transitions have been implicated as underlying important structure/function relationships. Figure 9 is essentially a redrawing of Figure 1, The Protein Trinity, with many more details included.

Databases of Intrinsic Order and Disorder

Given The Protein Trinity Hypothesis and the many examples of intrinsic disorder, it appears worthwhile to generate a database of proteins that have been characterized as intrinsically disordered. For purposes of comparison, it is useful to have a

database of intrinsically ordered protein segments in the same format.

Databases: Table 6 summarizes ordered and disordered data accumulated to date. To obtain a set of ordered protein segments, we started with a nonredundant set of proteins from PDB called PDB_Select_25.²⁴³ Sander et al. developed this dataset by grouping the proteins in PDB such that, within a group, every protein has >25% identity to at least one other member of the group, and between any two groups, no pair of proteins has >25% identity. Once the groups were determined, one member with the highest quality structure was chosen to represent each group.²⁴³ For each of the representative proteins, every disordered (unobserved) residue was removed, yielding the ordered dataset called *O_PDB_Select_25* (O signifies ordered). These ordered segments span all of the 3D structures known at the time this

Table 6. Summary of Databases of Intrinsic Order and Disorder

Database	Number of Segments	Number of Residues
O_PDB_Select_25	1 111	220 668
dis XRAY	56	2 844
dis NMR	41	4 019
dis CD	53	10 554
dis ALL	150	17 417
dis Fam32	572	52 688

database was constructed. The disordered sets are indicated as dis_X-ray, dis_NMR, and dis_CD according to the method used for the identification of disorder. The last group, dis_Fam32 was obtained by homology. A set of 32 proteins with disorder identified by a variety of means provided the starting points. Families of similar proteins were identified by sequence alignments using standard methods. Putative disorder in these proteins was then identified by alignment with the structurally characterized regions of disorder.

All of the disordered databases certainly contain segments having local tendencies for order. For example, segments with a high tendency for order are sometimes found to be binding sites within regions characterized as disordered.²⁴⁴ Likewise, the ordered databases certainly contain disordered segments misclassified as ordered. For example, several of the proteins in the ordered dataset are from co-crystals with DNA; such DNA-binding proteins often have significant regions of disorder when separated from their ligand. We are attempting to develop methods based on sequence analysis to identify ordered and disordered regions that are misclassified.

Comparison of Ordered and Disordered Protein Segments

Given the set of ordered and disordered amino acids, it is useful to compare them. We have carried out two comparisons. First, we compared the ordered and disordered segments with respect to their amino-acid compositions. Second, we compared these segments with respect to their sequence attribute values (e.g., hydropathy, hydrophobic moment, etc.).

Amino-acid compositions: To compare the compositions of the four disordered datasets with each other and with ordered data, we expressed the composition of each amino acid in a given disordered dataset as (Disordered-Ordered)/(Ordered). Thus, negative peaks indicate that disordered segments are depleted compared with the ordered ones in the indicated amino acids, and positive peaks indicate the reverse (Figure 10).

In Figure 10, the largest possible magnitude for a negative peak is -1 (e.g., for any amino acid that is completely missing from the given disordered region). In contrast, if a disordered region were composed exclusively of one amino acid, the positive peak value for that amino acid would be $(100 - \text{Ordered})/(\text{Ordered})$. For an amino acid that comprised 5% of the residues in the protein, the positive peak would have a value of 19. Despite this potential for a very large asymmetry, the positive and negative peaks in Figure 10 have similar magnitude, with positive peak values ranging from 0.025 to 0.61 and negative values from -0.007 to -0.70 .

The arrangement of the amino acids from least to most flexible in this figure was based on the scale established by Vihinen et al.,²⁴⁵ where their scale was defined by the average residue B-factors of the backbone atoms for 92 unrelated proteins. For the proteins in their study, the backbone atoms of W had the lowest B-factors on average and the backbone atoms of K had the highest. As the developers of this scale pointed out, the ranking does not reflect intrinsic flexibility, in which case G would have the highest rank. Rather, the ranking depends on the degree to which a given side chain tends to be buried (low ranking) or exposed (high ranking) in the crystal structure of globular proteins.

Overall, the four databases of intrinsic disorder are similar to each other for most of the amino acids.^{127,246} All four are substan-

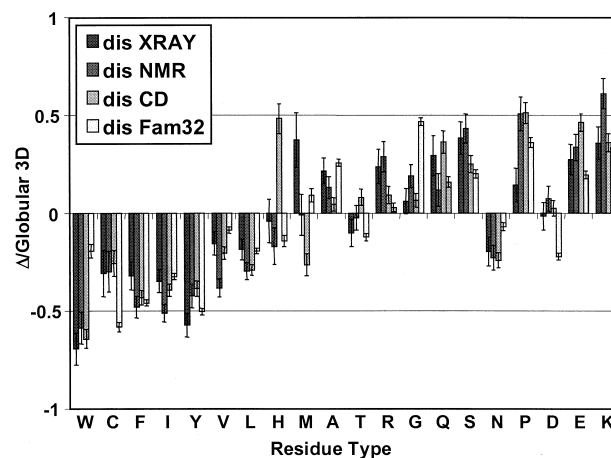


Figure 10. Amino-acid composition profiles. The amino-acid compositions of each disordered dataset in Table 6 are plotted as their differences from the compositions of the ordered dataset in this same table. Details are given in the text; briefly, a negative peak indicates that the given disorder dataset is depleted in that amino acid compared with the ordered set, while a positive peak indicates enrichment. The amino acids are arranged on the basis of B-factor estimates of residue flexibility, with the more rigid on the left and the more flexible on the right. Such B-factor estimates are not purely related to intrinsic flexibility but also include environmental effects.

tially depleted in W, C, F, I, Y, V, L, and N. We proposed that these be called order-promoting amino acids. All four are substantially enriched in A, R, G, Q, S, P, E, and K; we propose that these be called disorder-promoting amino acids. H, M, T, and D are not consistently enriched or depleted among the four sets of intrinsically disordered proteins. So these are neither order-promoting nor disorder-promoting.

T, N, and D stand out. All three are in a disorder-promoting neighborhood on the flexibility scale, yet T and N clearly promote order, not disorder, and D is ambivalent. These amino acids contain polar groups attached to the β -carbon and can readily form strong hydrogen bonds with the backbone. This side-chain-backbone interaction could lower the entropy of the disordered state, thus increasing the order-promoting tendencies of these residues compared with their neighbors in Figure 10.

Attributes: In addition to amino-acid composition, the disordered segments have also been compared with the ordered ones by various attributes such as hydropathy, net charge, flexibility index, helix propensities, strand propensities, and compositions for groups of amino acids such as W + Y + F (aromaticity). The values for these attributes were estimated by simple averages over windows of 21 amino acids. The disordered data was balanced by an equal amount of randomly selected ordered data, and then conditional probability plots were constructed (Figure 11). The ability of the various attributes to distinguish disorder from order can be compared by dividing the area between the two curves by the total area of the graphs, giving an area ratio (AR) value.²⁴⁷

Attribute ranking: Using the area-ratio method to assess the ability to discriminate order and disorder, we have ranked 265

property-based attribute scales²⁴⁶ plus more than 6,000 composition-based attributes²⁴⁸ (e.g., all possible combinations having one to four amino acids in the group). Many of the 265 scales and amino acids groupings are very highly correlated with each other. To find a set of such attributes having the best overall ability to distinguish between order and disorder, one needs to find a set of attributes that might not be the best individually, but that work best in combination. We have investigated various methods for obtaining optimal combinations of attributes, such as feedforward selection, branch and bound, and principal components analysis. Our previous studies^{126,249–252} were carried out using much smaller sets of disorder than those we recently accumulated. Studies are in progress to find such optimal sets for the current, much larger data sets of disorder. Presented in Table 7 is a representative, ranked set of attributes that provide fairly good discrimination between order and disorder.

Commonness of Disordered Regions as Estimated by Sequence Analysis

From the current set of experimentally characterized proteins with disorder, experimental biases preclude any meaningful estimate of the commonness of intrinsic disorder and of the relative amounts of the random coil and molten globule varieties. PDB is biased against disordered protein, mainly because of two factors:

1. At the protein purification step, intrinsic disorder could be underrepresented due to hypersensitivity to protease digestion.

Table 7. Ranking Properties for O/D Discrimination

Attribute	Area Ratio
14 Å contact number	0.542
Hydropathy ²⁸²	0.535
Flexibility ²⁴⁵	0.521
β-sheet	0.514
Coordination number	0.478
Hydropathy ²⁸³	0.421
RESP	0.334
Bulkiness	0.314
CFYW	0.285
Volume ²⁸⁴	0.246
Refractivity	0.242
Net Charge	0.236

2. Once purified, proteins with significant fractions of disorder don't crystallize. On the other hand, as discussed above, NMR is biased against molten globular disorder.

Finally, although use of both near- and far-UV CD in combination differentiates between random-coil and molten-globular disorder, the failure to routinely use near-UV CD means that, in practice, disorder identified by CD is almost all of the random coil variety. Structural characterization is much more tedious and expensive than determination of amino acid sequence. Furthermore, as discussed above, the structural char-

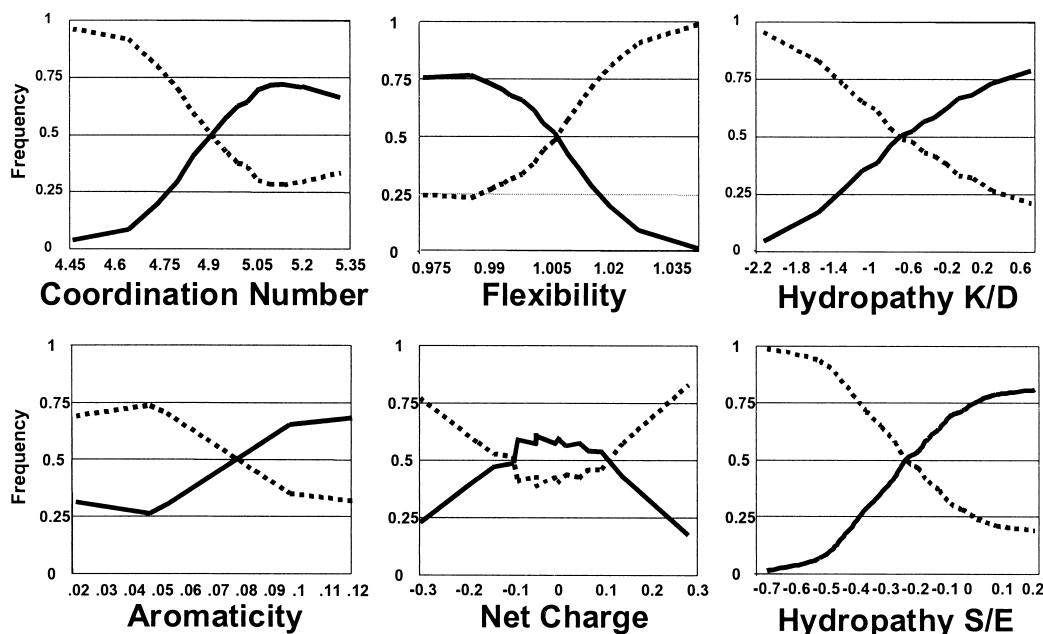


Figure 11. Sequence attributes. Balanced datasets of ordered and disordered residues were used to construct conditional probability curves that indicate the ability of a given attribute to discriminate between order (solid line) and disorder (dashed line). The greater the separation between the two curves, the better the discrimination. The coordination number relates to how many side chains can pack around a given side chain. Flexibility is the scale used for Figure 10. The two hydropathy scales are from Kyte and Doolittle²⁸³ (K/D) and from Sweet and Eisenberg²⁸² (S/E). Aromaticity is simply the sum of W + Y + F in a given region. Net Charge is simply (K + R) minus (D + E); including H had essentially no effect on these curves.

acterizations are biased. Thus, the most efficient and perhaps least-biased estimates of the commonness of intrinsic disorder are likely to come from sequence analysis (bioinformatics) methods as discussed below.

Computational methods for estimating intrinsic disorder or nonglobularity: Below we will discuss two methods: the widely used SEG analysis by Wootton et al.^{253–256} and our own efforts to develop predictors of disorder from amino acid sequence.^{126,248–252}

Sequence complexity using Shannon's entropy: Both Karlin^{257,258} and Wootton^{253,254} realized that some protein segments have statistically significant deviations from the average amino-acid composition, with fewer than the typical number of amino acids. Wootton et al. paid attention to protein structural characteristics of the unusual sequences, while Karlin et al. focused on the statistical aspects. We are following the approach of Wootton et al. because of their efforts to relate low-complexity sequences to protein structure.

Specifically, Wootton et al. recognized that fibrous proteins have fewer amino acids per unit length of sequence than do ordered, globular proteins, and they suggested that low-complexity sequences would be nonglobular if not fibrous. To identify such sequences, they proposed using Shannon's entropy, $K2$, which is given by $K2 = \sum (n_i/L) \log_2 (n_i/L)$, where the summation is over the 20 amino acids and where $n_i = 1$ if the i th amino acid is present in a window of length L and $n_i = 0$ if the i th amino acid is absent.

For windows of any length, the lower limit for $K2$ is 0, corresponding to 1 amino-acid type at every position in the window. For an alphabet of 20 characters and windows with $L \leq 20$, the upper limit increases for longer windows, reaching a plateau at ~ 4.32 for $L \geq 20$. Using this measure, silks, collagens, and coiled coils exhibit lower complexity values than an ordered database of protein segments derived from PDB (Figure 12), where all nonglobular and disordered regions have been removed. This set of ordered segments is called Globular-3D.

For complexity analysis, short windows exhibit large statistical variations and very long windows converge to the average complexity of the database. Our group, and Wootton et al. before us, using different methods of analysis, both found that windows of 45 residues provide a reasonable compromise between the two extremes.

Examination of over 2.5×10^6 windows of 45-residue ordered segments²⁵⁹ shows that no currently known ordered segment of this length from a globular protein contains fewer than ~ 10 different amino acids nor a $K2$ lower than ~ 2.9 . Furthermore, using laboratory evolution to select low-complexity sequences of the SH3 domain while maintaining its binding function as evidenced by its selection, Baker et al. developed a highly simplified version of this protein.²⁶⁰ The complexity ranges of wild type SH3 and of simplified SH3 are indicated in Figure 12 as horizontal bars, showing that the laboratory evolution experiments succeeded in markedly simplifying this protein. Nevertheless, the simplified protein has nearly identical lower bounds for number of amino acids and sequence complexity compared with the 2.5×10^6 windows for the current set of all known ordered proteins. The agreement between the laboratory selection and the sampling of natural selection suggests that about 10 amino acids and a $K2$

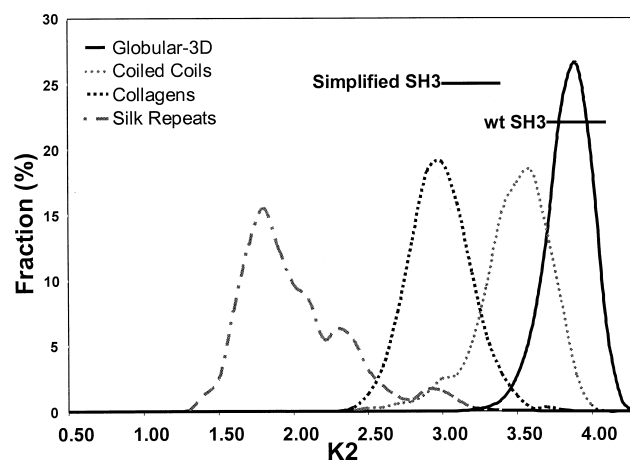


Figure 12. Sequence complexities. Smoothed histograms of $K2$ values for 4 sets of proteins: silks, collagens, coiled coils, and globular proteins. The identities of the proteins in each of these sets can be found at our website: www.disorder.wsu.edu.

of about 2.9 may represent the lower bound for the complexity of globular, ordered protein structure.²⁵⁹ Ordered helical bundles have been constructed with even lower sequence complexities, but these in our opinion are more like short segments of coiled coil rather than like true globular proteins.

Application of complexity analysis to the Swiss Protein database of protein sequences shows about 7% of all sequences have 45-residue windows with $K2$ values lower than 2.9 compared with 0.0% for ordered protein structure.¹²⁷ Furthermore, only a small fraction of these have sequence periodicities appropriate for one of the fibrous protein structures. Thus, there exists a class of almost certainly disordered protein structures having very low sequence complexity. We will show below that there is another class of disordered segments having high complexity. We speculate that sequences with low entropy disorder may loosely correlate with random coils and that sequences with high entropy disorder with molten globules.

Prediction of Disorder: As another means for estimating the commonness of intrinsically disordered proteins and understanding their functions, we are developing predictors of natural disordered regions (PONDRs). As shown in Figure 13, predictor development and application consist of several steps:

1. Construct databases of order and disorder for training.
2. Compare sequence attributes of ordered and disordered segments to find those to be used as inputs for the predictors.
3. Partition ordered and disordered data into training and test sets.
4. Train the predictors using the training sets and test using the test sets (e.g., cross validation).
5. Validate the predictors on additional ordered and disordered data.
6. Apply the predictors to single protein sequences, to databases of both structure and sequence, and to genomic sets of sequences.

PONDRs: Given sets of ordered and disordered segments, we have tried both linear (logistic regression and discriminant analysis) and nonlinear (neural network) models as

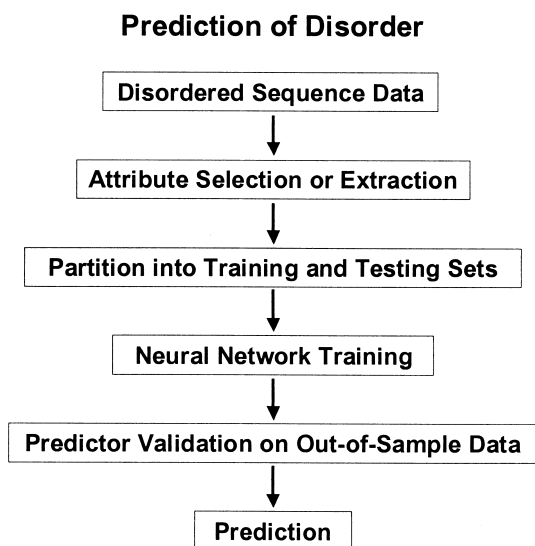


Figure 13. Steps for the development and application of PONDRs. The development and application of our predictors of natural protein disorder is shown schematically in this figure.

the basis for predictors of order and disorder. The nonlinear models are just slightly better than the linear ones. In our original work,¹²⁶ 10 attributes were selected for predictor development. The ordered and disordered data were randomly divided into five ordered and disordered subsets of equal size. Predictor training was carried out on 4/5 of the data and testing on 1/5. Overprediction was avoided by dividing the 4/5 set into a training set (80% of the 4/5) and a validation set (the remaining 20%). When testing accuracy on the validation set stopped increasing significantly, training was stopped. It was then that testing was performed on the testing set. The data was then recombined and a new predictor was trained to the same level of accuracy. This original predictor is herein called PONDR XL1 (X signifies X-ray characterized disorder; L signifies long regions of disorder).

A more recent predictor was developed using similar methods but included a larger training set, with examples characterized by various methods.¹²⁷ This predictor is called PONDR VL1 (V signifies variously characterized; L signifies long regions of disorder). Since we used windows of 21 and then smoothed the outputs by averaging over windows of nine, our initial predictor failed to give outputs for the first and last 14 amino acids in a given sequence.¹²⁶ Therefore, we developed two predictors, one using training data from the amino-terminal region and one from the carboxyl-terminal regions.²⁵¹ These end-specific predictors, herein called XT (X for X-ray characterized data, and T for their location at the termini) were later merged with the more recent internal-regions predictor to give our current PONDR VL-XT that has outputs from the first to the last residue in a sequence.¹²⁷

PONDR Accuracies: The datasets listed in Table 6 have not been studied for possible misclassification of ordered sequences as disordered. For the PONDRs developed to date, efforts were made to use disordered regions verified by more than one method and thereby reduce the amount of noise from

misclassification. The disadvantage of such an approach is that the training sets have been small.

Table 8 gives the accuracies of the various PONDRs on several of the ordered and disordered databases. As mentioned above, PONDR XL1 used only X-ray-characterized disorder and the training set was very small (only about 500 disordered amino acids),¹²⁶ while PONDR VL1 used X-ray- and NMR-characterized disorder and the training set was larger but still relatively small (about 1,300 disordered amino acids).¹²⁷ PONDR VL-XT involves a merger of the VL1 predictor with specific predictors for the two ends of the proteins.²⁵¹ For this predictor, two training accuracies are given. The value on the left is an average of the accuracies estimated for the first and last 10 residues; the value on the right is for the VL1 predictor. Finally, PONDR CaN was trained using the disordered regions from 12 calcineurin proteins.²⁴⁹

Note that all of the predictors exhibit similar accuracies on the ordered test set as observed during the training where 5-cross validation was used to estimate the accuracies. For the original XL1 predictor, the training accuracies ($73 \pm 4\%$) are close to the O_PDB_S25 values (68%); for the three other predictors, the corresponding accuracies are all above 80%. The improved accuracy for VL1 compared with XL1 probably relates to the increased size of the training set and the use of a better set of inputs.

Although the predictor accuracies on a large dataset of order matches the accuracies during training fairly well, there is a large drop in accuracy for each of the predictors when applied to out-of-sample disordered data. Our interpretation of these results is that all types of order, whether helices, strands, turns, or aperiodic, are fairly similar to each other, whereas the disordered regions show a much larger variation. Thus, a predictor trained on one type of disorder does poorly when applied to another type.²⁵² This point is discussed more fully below in terms of a specific prediction example.

We are currently attempting to classify the various types of disorder into groups, or “flavors,” and then construct order/disorder predictors for each group. Although preliminary, this approach appears to have promise. Partitioning our disordered set into three flavors leads to prediction accuracies above 80% for both order and disorder within a given flavor, but with very low out-of-flavor accuracies (as expected). For an unknown protein, its membership in one of the three flavors can be established by comparing amino-acid compositions.

Application to single proteins: One use is to PONDR single sequences, as shown in Figure 14A for the prion amino-

Table 8. PONDR Accuracies (% correct)

	Predictors			
	XL1	VL1	VL-XT	CaN
Residues	930	2 298	8 086	1 720
Training	$73 \pm 4\%$	$83.5 \pm 2\%$	$75.3\text{--}83.5\%$	$83 \pm 5\%$
O_PDB_S25	68	82	80	83
D_PDB_S25	49	44	63	33
NMR	52	59	64	31
CD	49	50	54	39
CaN Test	86	78	80	85

acid sequence and in Figure 14B for 4E-BP1. For prion, two PONDRs were used, one based on a training set containing a collection of different proteins, e.g., VL-XT, and one based on a training set containing long disordered regions from one family, calcineurin, with the disordered regions identified by homology (called CaN). For 4E-BP1, three PONDRs were used, VL-XT, CaN, and XL1.

Although PONDR CaN does as well as PONDR VL-XT for some proteins, for prion the CaN predictor misses the disordered region entirely. From several studies of this type, it is becoming clear that there are indeed a variety of different flavors of disorder.^{244,252} Much of our current work is devoted to determining methods to classify disordered regions into flavors, developing predictors for each flavor, and using such

predictors in attempts to understand flavor–function relationships.

Application to 4E-BP1 reveals predictions of order for all three PONDRs, where the predicted region of order is within a region characterized by NMR to be disordered (Figure 14B). The region predicted to be ordered by all three predictors corresponds approximately to a region of this protein responsible for binding to elongation factor EIF4E as indicated by the bar in the figure. As discussed above, this region undergoes a disorder-to-order transition upon binding.^{176,177} PONDR identified a binding site in 4E-BP1 by predicting order within a region structurally characterized to be disordered. This has been observed for a few other proteins as well, but the data are currently too limited to propose this result as a generality.²⁵⁰ Much more work is needed, but the prediction of order within a long, structurally characterized region of disorder has the promise of being a signal for the presence of a binding region that is independent of any particular sequence motif.

Commonness of disorder and low complexity: Although our predictors have a 20% error rate in the prediction of disorder for an ordered residue, the error rate drops significantly, to less than 0.4% for 40 consecutive predictions of disorder on segments known to be ordered. Thus, we are focusing our initial attention on such long disordered regions. To estimate the commonness of disordered regions of length $L \geq 40$ residues, we applied PONDR VL-XT to a nonredundant set of PDB proteins and to the sequences in Swiss Protein. Sequence complexity analysis was also applied to these same sets of proteins to find segments with $K2 \leq 2.9$. The results are shown in Table 9. For both data sets, there are fewer low-complexity segments than segments predicted as disordered and nearly all the low-complexity segments are also predicted as disordered.

Application of PONDR VL-XT suggests that 11% of the sequences in PDB_S_25 contain disordered segments. This value of 11% is much higher than the actual percentage of sequences in this structural database having disordered regions of $L \geq 40$ residues, suggesting that many of these predictions are errors. However, further examination shows that many of the apparent prediction errors are from segments that are involved with binding to DNA or to some other ligand. Such segments might undergo disorder-to-order transitions upon binding, in which case the prediction of disorder is correct, not an error. Thus, in the end, it is difficult to give an accurate estimate of errors in this analysis.

The current analysis predicts 29% of the sequences from Swiss Protein to contain at least one disordered region of 40 residues compared with 11% for PDB. This difference is so large that, despite the various uncertainties, it is safe to conclude that PDB is a biased set of proteins with fewer regions of disorder compared with the proteins in nature as represented by Swiss Protein. The obvious likely explanation is that the filter imposed by crystallization leads to exclusion of proteins with significant amounts of intrinsic disorder from the PDB. The protein with the largest amount of intrinsic disorder per unit cell in the current PDB is Bcl-x_L, with ~30% of this molecule being unobserved. Incidentally, the structure of this protein has been determined by both NMR and X-ray crystallography, with good overall agreement between the two methods in the assignment of disorder.³⁸

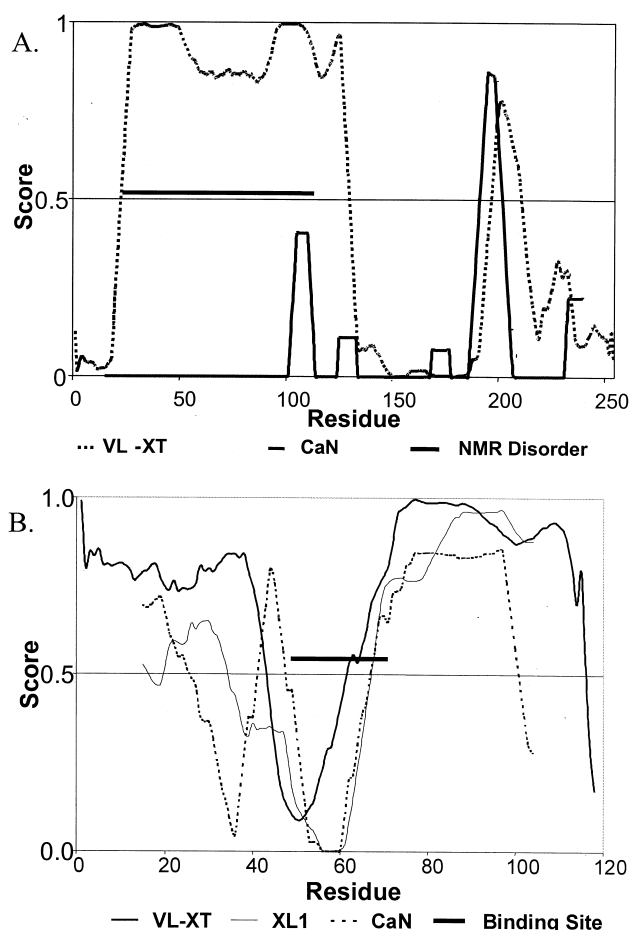


Figure 14. Application of various PONDRs to single sequences. In panel A, two PONDRs as indicated were applied to the mouse prion sequence and in B, three PONDRs as indicated were applied to 4E-BP1. For both panels, the ordinate shows the predictor output values (normalized to lie between 0 and 1) while the abscissa gives the residue number. Using the back-propagation learning algorithm, the PONDRs were trained to predict disorder by finding the weights that maximized the number of training-set disordered residues giving outputs > 0.5 while simultaneously maximizing the number of training-set ordered residues giving outputs < 0.5 .

Table 9. Commonness of Intrinsic Disorder

Category*	PDB_Select_25 No. chains	PDB_Select_25 % chains	SwissProt No. chains	SwissProt % chains
Proteins >45 aa	920	—	81 005	—
Low K2	6	0.7	5 748	7.1
Predicted LDRs	101	11.0	23 570	29.1
Extreme LDRs	5	0.5	6 291	7.8
Low K2, no LDR	2	0.2	881	1.1
LDR, no Low K2	97	10.5	18 703	23.1
Low K2, or LDR	103	11.2	24 451	30.2

* Chains having at least 1 of the indicated category.

Genomic Disorder

Probably the best way to estimate the commonness of intrinsic disorder is to predict disorder for the amino acids of whole genomes, including both the known and putative protein sequences. Such predictions could then be verified by laboratory experimentation.

We have carried out such predictions on 31 genomes, most

of which are complete (Table 10). These data used two measures of disorder. The first measure of disorder was the percentage of sequences in each genome with segments predicted by PONDR VL-XT to have ≥ 50 consecutive disordered residues. The second measure was the percentage of sequences in each genome to be wholly disordered, using a method based on cumulative distribution functions described in more detail elsewhere.²⁶¹

Table 10. Genomic Disorder

Kingdom	Species	# seqs	Length ≥ 50		CDF*	
Archaea	<i>Methanococcus jannaschii</i>	1 714	71	4%	26	2%
Archaea	<i>Pyrococcus horikoshii</i>	2 062	164	8%	70	3%
Archaea	<i>Pyrococcus abyssi</i>	1 764	157	9%	62	4%
Archaea	<i>Archaeoglobus fulgidus</i>	2 402	244	10%	93	4%
Archaea	<i>Methanobacterium thermoautotrophicum</i>	1 869	365	20%	140	7%
Archaea	<i>Aeropyrum pernix K1</i>	2 694	637	24%	490	18%
Eubacteria	<i>Ureaplasma urealyticum</i>	611	14	2%	9	1%
Eubacteria	<i>Rickettsia prowazekii</i>	834	23	3%	5	1%
Eubacteria	<i>Borrelia burgdorferi</i>	845	26	3%	14	2%
Eubacteria	<i>Campylobacter jejuni</i>	2 309	80	3%	21	1%
Eubacteria	<i>Mycoplasma genitalium</i>	480	20	4%	10	2%
Eubacteria	<i>Helicobacter pylori</i>	1 532	69	5%	24	2%
Eubacteria	<i>Aquifex aeolicus</i>	1 522	94	6%	29	2%
Eubacteria	<i>Haemophilus influenzae</i>	1 708	126	7%	27	2%
Eubacteria	<i>Bacillus subtilis</i>	4 093	323	8%	87	2%
Eubacteria	<i>Escherichia coli</i>	4 281	363	8%	107	2%
Eubacteria	<i>Vibrio cholerae</i>	3 815	333	9%	93	2%
Eubacteria	<i>Mycoplasma pneumoniae</i>	675	60	9%	14	2%
Eubacteria	<i>Xylella fastidiosa</i>	2 761	246	9%	103	4%
Eubacteria	<i>Thermotoga maritima</i>	1 842	165	9%	53	3%
Eubacteria	<i>Neisseria meningitidis MC58</i>	2 105	190	9%	64	3%
Eubacteria	<i>Chlamydia pneumoniae</i>	1 052	100	10%	40	4%
Eubacteria	<i>Synechocystis sp</i>	3 167	338	11%	104	3%
Eubacteria	<i>Chlamydia trachomatis</i>	894	99	11%	42	5%
Eubacteria	<i>Treponema pallidum</i>	1 028	115	11%	37	4%
Eubacteria	<i>Mycobacterium tuberculosis</i>	3 916	747	19%	293	7%
Eubacteria	<i>Deinococcus radiodurans chr I</i>	2 580	534	21%	212	8%
Eucaryote	<i>Caenorhabditis elegans</i>	17 049	4 636	27%	1 322	8%
Eucaryote	<i>Arabidopsis thaliana chrII, IV</i>	7 849	2 248	29%	653	8%
Eucaryote	<i>Saccharomyces cerevisiae</i>	6 264	1 858	30%	356	6%
Eucaryote	<i>Drosophila melanogaster</i>	13 885	5 651	41%	2 403	17%

* Sequences in each genome judged to be wholly disordered, using a method based on cumulative distribution functions.²⁶¹

To our surprise, the amount of predicted disorder shows a wide range among organisms from the 3 kingdoms, with the eucarya exhibiting more disorder by these two measures than either the procarya or the archaea. For the first measure, the four eucarya were predicted to have 27–41% (!) of their proteins with disordered regions of length ≥ 50 consecutive residues. As for the second measure, the four eucarya were predicted to have ~6–17% of their chains to be wholly disordered like FlgM, 4E-BP1, HMG-I(Y), or neurogranin/neuromodulin. *Drosophila melanogaster* appears to have much more disorder than the other 3 eucarya. This may be due to the incomplete nature of the *Drosophila* database, or to the physiology of an organism that undergoes several developmental stages.

Neither of these estimates is corrected for false-negative or false-positive predictions. From Table 8, PONDR VL-XT seems to yield much larger false negative than false-positive predictions. If this trend carries over to predictions on genomic sequences, then the 30% and 7% values for proteins with disordered regions of length ≥ 50 and for proteins that are wholly disordered, respectively, are substantial underestimates. These data suggest that intrinsic disorder is, indeed, a very common element of protein structure.

When considering a region of intrinsic disorder of length ≥ 50 , Figure 6C should be kept in mind. The calsequestrin monomer in this figure has 367 amino acids, yet an extended disordered tail of just 20 residues more than crosses the entire diameter of the protein. An unfolded disordered segment of 50 residues would be long enough to occlude a very large fraction of the surface of a protein of this size.

Disorder and Intracellular Protease Digestion

Sensitivity to protease digestion is sometimes used as an argument against the existence of disordered proteins *in vivo*. To the contrary, there are so many cellular events controlled by protease digestion that intrinsic disorder is almost certainly a crucial (but, for the most part, unrecognized) component of proteolytic regulation.

Cells regulate their resident proteins through synthesis and degradation; together these two processes are called turnover. Schoenheimer²⁶² first recognized this dynamic behavior in 1942. In eucaryotic cells, individual proteins show characteristic half-lives, and for different intracellular proteins these half-lives vary over a wide range, from a few minutes to twenty days or more.²⁶³ Studies such as these suggest a great deal of control and regulation.

A number of specific mechanisms for regulating intracellular proteolysis have come to light in recent years. A few examples include: regulation by the presence of certain sequences, such as regions rich in PEST;^{264,265} regulation by the ubiquitin/proteasome system for digestion of unfolded protein;^{266,267} and regulation by calcium levels via the calpain/calpastatin system.^{268–270}

Specific protease digestion events utilizing one or more of the protease systems mentioned above are linked by a very large number of regulatory processes. To give a small sample from a very long list, events regulated by protease digestion include the well-known activation of various inactive precursors including peptide hormones,²⁷¹ the control of cell cycle progression,^{272,273} the activation of programmed cell death,^{207,274} and even the regulation of cholesterol content.²⁷⁵

Given the importance of flexibility for protease diges-

tion,^{88,276} we suggest another obvious point of control for proteolytic regulation, namely by modulating the order/disorder status of the protease sensitive site. Such modulation could be accomplished by phosphorylation/dephosphorylation, protein–ligand interactions, and protein–protein interactions, for example. With respect to the last possibility, the current view is that chaperone proteins exist for the purpose of mediating protein folding into the correct 3D structure.^{277,278} From the point of view of intrinsic disorder, the true functions of some of these proteins might be to modulate accessibility to loci important for protease control. For example, calmodulin is substantially disordered in the absence of calcium and so might be expected to be digested via the ubiquitin/proteasome system. However, as discussed above, a collection of proteins with IQ motifs bind to calmodulin at low calcium. Perhaps the function of this association is not only for storage as was suggested previously¹⁸⁷ but also to protect the unfolded calmodulin from protease digestion.

Implications for Structural Genomics and Proteomics

The first grants to support consortia for high throughput protein structure determination have now been awarded, thus marking the actual start of structural genomics.^{279,280} This effort is squarely based on the paradigm: *Amino-Acid Sequence* $\rightarrow$ *3D Structure* $\rightarrow$ *Function*. Of course the goal of these efforts is to use structural information to help catalog the functions of the protein in the human and other genomes. Given the information and discussions provided in this review, which are from a similar perspective as information and discussions published previously,²⁸¹ it is evident that the structural genomics project will be very seriously incomplete unless a more systematic effort is made to discover and study intrinsically disordered proteins.

SUMMARY

Intrinsic protein disorder is very likely encoded by the amino-acid sequence and represents a common feature of native proteins. Thus, disorder ought to be recognized as a distinct category of protein structure. A protein with intrinsic disorder can bind permanently with a partner and thereby exist as a structured protein throughout its lifetime. Proteins with intrinsic disorder can undergo disorder-to-order transitions upon binding with one or more partners and thus be structured part of the time and unstructured part of the time. Finally, proteins with intrinsic disorder can in some cases carry out function without ever becoming ordered and thus remain disordered throughout their existence. To account for all of these possibilities, a new paradigm for protein structure/function is needed; we therefore propose The Protein Trinity as illustrated and explained in Figure 1.

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and Walter Englander are thanked for directing us to the origins of the primary-secondary-tertiary hierarchy for protein structure. Next, Charles DeLisi and coworkers Sandor Vajda, Rakefet Rosenfeld, and, especially, Zhiping Wang, are thanked for helping us avoid mistakes in the interpretation of their very important paper on the energetics of flexible ligand binding. Ya-Yue Van is acknowledged for naming The Protein Trinity. Finally, we want to thank NIH, NSF, DOE, and the MRC of Canada for the various grants that enabled the authors to meet and collaborate on the study of intrinsic protein disorder, and especially for NIH-R01-LM06916, NSF-CSE-II-9711532, and NSF-REU-IID-9711532, which were awarded expressly for the purpose of studying intrinsic disorder.

Note added in proof

In the discussion of The Protein Trinity herein, the molten globule to random coil transition is said to be non-cooperative. It has come to our attention that non-cooperative unfolding is not a universal trait for all molten globules. One molten globule, apomyoglobin, does exhibit a slight degree of cooperativity upon unfolding, although the molten globule is much more fragile than the ordered state.²⁸⁵ In this same section, we also mention that fully unfolded proteins are generally not true random coils, but we use this terminology for simplicity. A recent discussion of the relationship between unfolded protein and the random coil is quite helpful.²⁸⁶

The studies indicating that nucleosome hyperacetylation reduces particle rigidity have been reported in preliminary form,²⁸⁷ as have the studies indicating that clusterin appears to contain a native molten globule that may function as a proteinaceous detergent.²⁸⁸

Finally, after this paper went to the press, a paper was published indicating that hydropathy and net charge could be used in combination to identify “natively unfolded proteins”.²⁸⁹ By “natively unfolded” the authors mean lack of any appreciable folding (Uversky, personal communication). Perhaps combining our predictors with those of Uversky et al. would provide a means to indicate protein domains that are likely to be native molten globules. Let D be the set of all disordered proteins as approximately identified by our predictors. Let RC be the set of all native random coils as approximately identified by the predictor of Uversky et al.,²⁸⁹ and RC' be the complement of this set. We speculate that putative native molten globules, MG , could then be identified as $MG \equiv D \cap RC'$, or the set of all disordered proteins from which all random coils have been removed.

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